Algebra as a structural toolkit

Foundations, source comparisons, and a Hadamard/Williamson bridge

Research notes for returning to algebra

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1 Returning to algebra: a compact working guide

1.1 The essential memory spine

Retain the following statements with their hypotheses and the type of every map. The rest of the toolkit should be reconstructible from them.

1. A map $V/U \to X$ is a map $V \to X$ that kills U.
2. Dualization is precomposition; evaluation gives $V \cong V^{**}$ in finite dimension. An identification with V^* needs extra data.
3. A direct sum is encoded by projections, and $P^2 = P$ gives $V = \operatorname{im} P \oplus \ker P$.
4. A split exact sequence has a section. Sections differ by a map from the quotient to the kernel; existence does not imply a preferred choice.
5. An operator is a $k[t]$-module. The PID structure theorem classifies finite-dimensional operators through invariant factors; Jordan form needs splitting of the minimal polynomial.
6. Coprime annihilator factors produce polynomial idempotents by Bézout.
7. Tensor products represent bilinear or balanced maps. A formula on pure tensors must respect the defining relations.
8. $W \otimes V^* \cong \operatorname{Hom}(V, W)$ when V is finite-dimensional. Trace is evaluation; not every infinite-rank operator is an algebraic tensor.
9. Tensor, symmetric and exterior algebras are free constructions with specified relations. Λ^r transports alternating data; the top exterior power gives a determinant line.
10. Forms transform by congruence, operators by similarity. A positive metric supplies adjoints, orthogonal projections, and the normal spectral theorem.
11. Exactness is image equals kernel. Over a field short exact sequences split and positive Ext/Tor vanish; homology of an arbitrary complex need not.
12. A representation is a module over a group algebra. Maschke needs invertible group order; Schur's scalar conclusion needs finite dimension and an algebraically closed field.

For Hadamard/Williamson add five facts: Gram orthogonality is an integral identity; tensors multiply valid orders; cyclic Fourier coordinates are CRT coordinates; compression loses lifting information; and equality of Boolean phases must be distinguished from independence of their signs.

1.2 A selective route, with stopping tests

First return: regain the language. Read the quotient/duality/splitting chapter, with Xi II §§1.4–1.6 and 2.4 beside it. Stop when you can construct the quotient map and its dual, describe the affine space of sections, and explain why an invariant complement is an additional demand. Xi I can be a reference for matrix maps and determinant characterizations rather than a sequential restart.

Second return: recover operator theory from one theorem. Choose Grillet VIII.6–7 or Hungerford IV.6 then VII.4–5 as the principal module route. Read Xi II §2.5's actual Bézout and quotient-induction proofs as a comparison, not a second complete linear algebra course. Stop when you can obtain primary projections, invariant factors, Jordan blocks under the field hypothesis, and the block counts from kernel growth.

Third return: turn multilinearity into maps. Read Xi II §6.2, especially Theorem 6.15 and the Hom/scalar-extension examples. Then use Grillet XI.5–6 and XIII.2–4 for balanced products, quotient constructions, and characteristic issues. Read Mac Lane III.1–2 and IV.1 when the universal examples are already familiar. Stop when you can define tensor-induced maps without a basis and derive Cauchy–Binet from $\Lambda^r(gf)$.

Keep a metric branch active. Read Xi II §§1.7, 3.3–3.5 and 3.8 as needed for forms, adjoints, normal operators, and projections. This is central for Hadamard Gram matrices. Defer the rest of affine/projective geometry only according to current goals, without concluding that the content was replaced.

Introduce homology at its first real obstruction. Read the exactness chapter here, then Hilton–Stammbach III.1–4 and IV.1–3 if you want the general construction. Grillet X.1, X.3 and XII are an alternative. Stop the first pass after computing the $k[t]/(t^2)$ extension and explaining nonzero field-complex homology. Spectral sequences and derived categories are not prerequisites for the current Hadamard bridge.

For the confirmed problem direction. Read the Hadamard bridge in order: Gram matrices and tensor closure; Xi III §5.6 finite fields; the character-sum proof and Paley blocks; cyclic group algebra and Fourier CRT; then the pinned third-round proofs with their color and Boolean equations. Use Xi III Chapter 3 or Grillet IX.7 for group algebras and representations, keeping Grillet's initial indecomposable/irreducible convention visible. Read the prior limitations before proposing a larger search. The essential stopping test is being able to say exactly which implication is necessary, sufficient, or only a relaxation.

Hall's further Lie theory, Lang's local arithmetic, Hartshorne's scheme/sheaf theory, and Hirsch's transversality/bundle development can then be pursued as separate branches. Their prerequisite bridges are recorded, not placed in front of every use of linear algebra.

1.3 Reconstruction exercises

These are deliberately small. Their purpose is to test whether a tool can be rebuilt, not to begin the later difficult problem-set campaign.

1. **Quotient versus complement.** For $U \subseteq V$, construct $(V/U)^* \cong U^\circ$ and show why no complement is needed. Then describe all sections of $V \to V/U$ once one has been chosen.
2. **One invariant line.** In $k[t]/(t^2)$, list the subspace killed by t. Prove it has no t-invariant complement, and identify the associated nonzero Ext class.
3. **Recover a projection.** Assuming $T^2(T-I)^2 = 0$, derive the projector $3T^2 - 2T^3$ from polynomial congruences. Check the argument in characteristics 2 and 3 without dividing by those integers.
4. **Recover Jordan data.** An operator has only eigenvalue λ and successive kernel dimensions $0, 4, 7, 9, 10, 10$. Find all its block sizes, minimal polynomial, and the dimension of its commutant.
5. **Nonsplitting field.** Classify multiplication by t on $\mathbb{R}[t]/(t^2+1)$ as a real module and after complexification. Explain which use of "semisimple" is valid before extension.
6. **Tensor universality.** Construct the currying bijection and $W \otimes V^* \to \mathrm{Hom}(V, W)$, specifying precisely which one needs finite-dimensional V. Prove the finite-rank obstruction.
7. **Determinants from wedges.** Derive the minors formula for $\Lambda^2 f$ and the 2×2 Cauchy–Binet identity for a 2×3 matrix times a 3×2 matrix.
8. **Forms versus operators.** Give two congruent real symmetric matrices that are not similar. Derive $H^{-1}A^*H$ from the defining adjoint identity, stating the Hermitian linearity convention.
9. **Homology survives.** For $d : k^3 \to k^2$, $d(x, y, z) = (x, y)$, compute the two homology groups of the two-term complex, and write an explicit contractible/homology decomposition.
10. **The right group action.** Verify the inverse in the dual action, and differentiate it to obtain the minus sign in the Lie algebra dual action.

11. **Character sum.** Reproduce the count of $y^2 = x(x+a)$ over an odd finite field. Identify every use of $a \neq 0$ and $2 \neq 0$, then check one Paley block multiplication.
12. **Compression and phase dependence.** Check the core-nine compressed autocorrelation example. Explain why failure of its chosen lifts alone cannot prove absence of all lifts. Separately give a numerical assignment to three formally independent signs that violates $\xi_{p+q} = \xi_p \xi_q$.

1.4 Brief answer checks

For Exercise 1, pull back along the quotient, and use $s' = s + ih$. For 2, the killed subspace is kt and the class is $1 \in k$ under $\mathrm{Ext}^1_{k[t]}(k,k) \cong k$. For 3, require a polynomial to be zero modulo t^2 and one modulo $(t-1)^2$; the integer identity remains valid after any reduction of characteristic.

For 4, the differences are $4, 3, 2, 1, 0$, so the sizes are $4, 3, 2, 1$. The minimal polynomial is $(t-\lambda)^4$ and the commutant dimension is $\sum_{i,j} \min(i,j) = 30$ for $i, j = 1, 2, 3, 4$. For 5, the real module is simple; after complexification it is the direct sum of the i and $-i$ eigenspaces. It is semisimple over $\mathbb{R}[t]$ but not real diagonalizable.

For 6, currying needs no finite dimension, while surjectivity of the indicated Hom/tensor map does. For 7, sum the three intermediate 2-subsets. For 8, I and $2I$ suffice, and $h(x,y) = \bar{x}^{\mathsf{T}} H y$ is the convention used here. For 9, H_1 is the z-axis and $H_0 = 0$; the (x,y) coordinates map isomorphically to k^2. For 10, transpose reverses products and differentiation of an inverse at the identity contributes a minus sign. For 11, the count is $q-1$. For 12, choose all three phase signs negative: the product of the first two is positive, contradicting the third. These checks indicate the target, while the chapters provide the proofs to reconstruct.

1.5 What can be deferred

Defer a formula once its map or universal property can be reconstructed. Defer a subject when it has no role in the current destination. Do not defer the hypotheses, the equivalence relation, or a known failure case. For this collection, an effective memory sheet contains definitions and four or five counterexamples, with links back to proofs, rather than a long inventory of named theorems.

2 Architecture: structures, maps, and obstructions

2.1 The organizing claim

Much ordinary linear algebra is the study of finite-dimensional vector spaces with additional structure. The most reusable move is to identify that structure, declare which maps preserve it, and then ask which familiar constructions remain available. An abstract tool earns its place when it preserves information or exposes an obstruction that a coordinate calculation conceals.

Four layers organize the collection:

1. **Linear structure.** Kernels, images, quotients, duals, direct sums, and tensor products describe how linear information is discarded, tested, assembled, and transported.
2. **An action or a multiplication.** An endomorphism makes a vector space a $k[t]$-module. A group action makes it a $k[G]$-module. A bilinear multiplication is a map $V \otimes V \to V$ satisfying specified identities.
3. **Decomposition and its obstructions.** Idempotents encode direct sums; coprime annihilators produce idempotents; semisimplicity guarantees invariant complements. Extensions measure ways in which complements fail to exist.

4. **Variation and locality.** Families of vector spaces introduce bundles and sheaves; changing coefficients introduces localization, integrality, and arithmetic. A theorem about each individual vector space need not control a family or an integral lattice.

These layers are connected, but none erases the others. A classification up to arbitrary linear isomorphism does not solve a problem about lengths, positive forms, an integral lattice, or smooth variation.

2.2 The prerequisite map

The following arrows mean a useful prerequisite for the stated destination, not a claim that every possible approach must take this route.

Foundation	Construction or theorem it supports	Destination and extra bridge
Equivalence relations, linear maps, subspaces	Quotients, kernels, first isomorphism theorem	Exact sequences; then modules and abelian categories if needed
Bases, linear functionals, quotients	Dual exact sequences, annihilators, evaluation	Adjoints after choosing a form; trace duality after separability
Direct sums, polynomial division, Bézout	Coprime primary decomposition	$k[t]$-modules, rational/Jordan forms, commuting operators
PID ideals and presentations	Smith normal form, cyclic module decomposition	Integral abelian groups and operator classification; higher-dimensional polynomial rings need different tools
Bilinear maps and quotients	Tensor products and Hom–tensor adjunction	Scalar extension, representations, multilinear algebra
Tensor algebra and relations	Symmetric and exterior algebras	Polynomial coordinate rings, determinants, differential forms after smoothness
Splitting and projective presentations	Ext^1, Tor_1	Extension obstructions; derived functors after complexes and homotopies
Groups, invariant subspaces, duals, tensors	Group representations, Maschke, Schur	Matrix Lie groups after topology and the exponential map; Lie algebra representations after differentiation
Finite field extensions and multiplication maps	Norm, trace, separable trace pairing	Lang: integral closure, Dedekind ideals, valuations, completions, ramification
Commutative rings, modules, localization	Affine coordinate rings, local rings, tangent spaces	Hartshorne: sheaves, schemes, locally free sheaves, cohomology
Real linear maps, inner products, multivariable calculus	Tangent spaces, normal spaces, quadratic forms	Hirsch: manifolds, function-space topology, approximation, transversality, bundles

An economical order for this study is

$$\text{maps and quotients} \longrightarrow \begin{cases} \text{duality and splitting}, \\ \text{polynomials and modules}, \\ \text{bilinear maps and tensors}. \end{cases}$$

These three paths rejoin at invariant decompositions, determinant lines, forms, and representations. Exactness becomes useful as soon as a construction stops preserving injections or an invariant complement fails. There is no need to put spectral sequences in front of that first obstruction.

2.3 What the collection actually contributes

Xi's 基础代数 I develops the concrete languages of equations, matrix maps, determinants, polynomial division, roots, symmetric polynomials, and resultants. Volume II develops vector spaces and operators, but also the metric and geometric meaning of those objects. Volume III adds transformation groups, representations, modules, Lie algebra examples, and Galois theory. This is a connected course, not merely three fragments awaiting replacement by a graduate book.

Grillet is particularly well aligned with the present structural route: VIII.6–7 connects PID modules to Jordan form, X handles splitting and projectivity, XI develops Hom and tensors, XII introduces Ext/Tor, and XIII builds free algebras and their quotients. Hungerford independently recommends PID torsion modules before his linear algebra chapter (printed p. xviii, PDF p. 15). This is strong evidence that the module viewpoint is a sound organizational choice, rather than a gratuitous modernization of the user's plan.

Mac Lane supplies a language for recognizing the same construction in different categories. His universal arrows and representability are useful after several concrete examples have been worked through. They do not by themselves provide Jordan bases, a spectral theorem, or a treatment of Euclidean geometry.

Hilton–Stammbach supplies the next layer when exactness fails: projective presentations, extension classes, complexes, and derived functors. Its route through extensions before general derived functors is especially compatible with a reader coming from linear algebra. Hall turns familiar matrix groups and the exponential into systematic Lie theory and develops representation theory beyond the initial $\mathfrak{sl}_2$ example.

Hartshorne and Lang are destinations requiring new bridges. Neither should be mistaken for an interchangeable abstract-algebra text. Hirsch is a different destination again: his preface (printed p. vii, PDF p. 8) deliberately avoids tensors and differential forms, and largely avoids algebraic topology. The relevant foundations for his approach are geometry, analysis, and topology as well as linear algebra.

2.4 The memory hierarchy

There are three different things to retain.

Definitions that reconstruct maps. Remember the quotient factorization property, tensor universality, evaluation, and the equations defining a split sequence or an idempotent. Each tells you how a new map must be defined and what must be checked for it to exist.

Theorems whose hypotheses control classification. Remember finite generation and the PID hypothesis in the module structure theorem; splitting of the minimal polynomial for Jordan form; positivity and the base field in the spectral theorem; invertibility of $|G|$ in Maschke's theorem; separability in trace duality. The missing hypothesis often explains the entire difficulty.

Geometric or computational interpretations. A quotient is not a preferred complement. A dual is not an inner-product identification. A tensor is not usually a pure tensor. A canonical form

is not a canonical basis. A fiber is not a smooth bundle, and a finite-dimensional vector space over a number field is not its integral lattice. These distinctions protect the structural tools from misuse.

2.5 What this architecture does not promise

Two commuting operators give a module over $k[s,t]$, which is generally not a PID. The one-operator classification therefore does not automatically extend to pairs. Likewise, finite-dimensionality over k does not make a $k[G]$-module semisimple, and a bundle of rank n does not automatically possess a global frame. These are productive boundaries, not reasons to abandon abstraction.

The existing algebra direction has been confirmed as the Hadamard/Williamson study. Its dedicated bridge locates its Gram matrices, integral cyclic group algebras, finite-field characters, symmetry quotients, and Boolean constraints within this map. The other subject bridges retain their independent value; they are not presented as prerequisites for that particular problem.

3 Quotients, duality, and splitting

3.1 Conventions and the type of a map

Throughout the core, k is a field and vector spaces are finite-dimensional unless a paragraph explicitly removes that hypothesis. Rings are associative with identity, modules are unital, and modules are left modules unless sides are specified. A commutative ring is stated when tensor products are themselves used as modules over that ring. The zero vector space is allowed.

For a linear map $f : V \to W$, f is the map and $f(v)$ is its value at $v \in V$. Write $\operatorname{im} f \subseteq W$ and $\ker f \subseteq V$. Composition $g \circ f$ first applies f. An equality between maps includes their domains and codomains; when a subspace is identified with its image, the inclusion will be stated. This matters especially when comparing an operator on a quotient with a restriction to a complement.

3.2 Quotients: impose relations once

Let $U \subseteq V$ be a subspace. Define $v \sim w$ if $v - w \in U$ and $V/U = \{v + U : v \in V\}$. Addition and scalar multiplication of cosets are well defined because replacing representatives changes the result by an element of U. Let $q : V \to V/U$, $q(v) = v + U$.

Quotient principle. For any vector space X and linear map $f : V \to X$, there is a unique linear map $\bar{f} : V/U \to X$ with $\bar{f} \circ q = f$ if and only if $U \subseteq \ker f$.

Proof. Necessity follows from $q(u) = 0$ for $u \in U$. Conversely set $\bar{f}(v + U) = f(v)$. If $v - w \in U$, then $f(v) = f(w)$; hence this is well defined. It is linear by the coset operations. Surjectivity of q forces uniqueness. □

The construction has a concrete description, but its reusable content is the factorization property. When constructing a map from a quotient, specify its value on a representative and prove it annihilates the relations. When proving two maps from the quotient equal, precompose with q.

First isomorphism theorem. If $f : V \to W$ is linear, then

$$V/\ker f \xrightarrow{\tilde{f}} \operatorname{im} f, \qquad \tilde{f}(v + \ker f) = f(v)$$

is a canonical isomorphism. It is surjective by definition of the image; its kernel is zero by definition of $\ker f$. Thus rank–nullity follows from dimension of a quotient, rather than being an isolated formula.

Why universal objects are unique. If (Q, q) and (Q', q') have the same quotient property for $U \subseteq V$, there are unique maps $a : Q \to Q'$ and $b : Q' \to Q$ satisfying $aq = q'$ and $bq' = q$. Then $baq = q$, so uniqueness gives $ba = 1_Q$; similarly $ab = 1_{Q'}$. Uniqueness means a unique isomorphism preserving the specified structure, not a unique bare vector-space isomorphism.

Two useful reconstructions are

$$U/(U \cap W) \cong (U + W)/W, \qquad (V/U)/(W/U) \cong V/W \quad (U \subseteq W \subseteq V).$$

For the first, restrict the quotient map $V \to V/W$ to U and compute its kernel. For the second, send $(v + U) + (W/U)$ to $v + W$ and apply the quotient principle twice. These proofs explain the maps as well as the dimensions.

Recognition signal. Whenever a problem says "modulo," identifies choices, or asserts that a quantity depends only on an equivalence class, a quotient factorization is present. If an operation is not compatible with the relation, the quotient does not inherit that operation. For example, $T : V \to V$ induces $\bar{T} : V/U \to V/U$ precisely when $T(U) \subseteq U$.

3.3 Duality: test vectors by scalar-valued maps

Define $V^* = \mathrm{Hom}_k(V, k)$. For $f : V \to W$ define

$$f^* : W^* \to V^*, \qquad f^*(\lambda) = \lambda \circ f.$$

Then $(g \circ f)^* = f^* \circ g^*$ and $1_V^* = 1_{V^*}$. The reversal is forced by the domains. In dual bases the matrix of f^* is the transpose of the matrix of f; complex conjugation is absent here.

The evaluation map is

$$\epsilon_V : V \to V^{**}, \qquad \epsilon_V(v)(\lambda) = \lambda(v).$$

If $e_1, \dots, e_n$ is a basis with dual basis $e^1, \dots, e^n$, then $\epsilon_V(e_i)$ is the corresponding dual basis of V^{**}. Thus ϵ_V is an isomorphism. It is natural: for $f : V \to W$,

$$f^{**} \circ \epsilon_V = \epsilon_W \circ f.$$

Indeed, evaluate either side at $v \in V$ and $\lambda \in W^*$; both give $\lambda(f(v))$. The proof uses a basis to establish invertibility, but the map itself uses no chosen basis.

There is generally no such preferred identification $V \cong V^*$. A basis or a nondegenerate bilinear form provides one, and different choices provide different maps. Naturality of the double dual should not be confused with a metric on V.

For $U \subseteq V$ define its annihilator

$$U^\circ = \{\lambda \in V^* : \lambda(u) = 0 \text{ for every } u \in U\}.$$

The quotient principle identifies $(V/U)^*$ canonically with U°: pull back along q. Restriction to U gives an exact sequence

$$0 \longrightarrow (V/U)^* \xrightarrow{q^*} V^* \xrightarrow{\text{res}} U^* \longrightarrow 0.$$

To prove surjectivity of restriction, extend a basis of U to one of V, and prescribe zero on the added vectors. The extension exists; it is usually not unique. The sequence is canonical even though a chosen splitting is not.

It follows that $\dim U^\circ = \dim V - \dim U$. Under $V \cong V^{**}$, $(U^\circ)^\circ = U$: inclusion is immediate, and the dimensions agree. For $f : V \to W$,

$$\ker f^* = (\operatorname{im} f)^\circ, \qquad \operatorname{im} f^* = (\ker f)^\circ.$$

The second identity follows by factoring a functional vanishing on $\ker f$ through $\operatorname{im} f$ and extending it to W. Consequently $\operatorname{rank} f^* = \operatorname{rank} f$ and row rank equals column rank. These are manifestations of testing a map by functionals.

Boundary. For infinite-dimensional algebraic vector spaces, dualization of short exact sequences is still exact when bases and their usual choice principles are available. But $V \to V^{**}$ need not be surjective and $V^* \otimes W \to \operatorname{Hom}(V, W)$ generally has only finite-rank maps in its image. Continuous duals in analysis form a different theory; the algebraic extension argument need not preserve continuity.

3.4 Direct sums and idempotents

The external direct sum $A \oplus C$ has inclusions i_A, i_C and projections p_A, p_C satisfying

$$p_A i_A = 1_A, \quad p_C i_C = 1_C, \quad p_A i_C = p_C i_A = 0, \quad i_A p_A + i_C p_C = 1_{A\oplus C}.$$

These equations are more useful than the ordered-pair notation: they allow block matrices and maps to be reconstructed without coordinates. A finite direct sum is both a product and a coproduct. For infinite families, the direct sum consists of finitely supported tuples, whereas the product includes all tuples; they are different constructions.

Idempotent principle. For an endomorphism $P : V \to V$, the equation $P^2 = P$ is equivalent to a decomposition

$$V = \operatorname{im} P \oplus \ker P$$

in which P is projection onto the first summand.

Proof. Write $v = P(v) + (v - P(v))$; the second term lies in $\ker P$. If $x = P(y)$ and $P(x) = 0$, then $x = P^2(y) = P(y) = 0$. Conversely projection in a direct sum plainly satisfies $P^2 = P$. □

This works for modules over any ring. If P commutes with an operator T, both summands are T-invariant. Conversely a decomposition into T-invariant summands gives a projection commuting with T. Thus finding a decomposition can be translated into finding idempotents in a commutant.

Failure of a tempting test. Pairwise zero intersection of three subspaces does not imply their sum is direct. In k^2, the lines spanned by e_1, e_2, and $e_1 + e_2$ have pairwise intersection zero, but satisfy a nontrivial three-term relation. The right test is that the addition map $\bigoplus_i U_i \to V$ be injective, or that $U_i \cap \sum_{j \neq i} U_j = 0$ for each i.

3.5 Splitting: existence versus choice

A short exact sequence

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

means that i is injective, p is surjective, and $\ker p = \operatorname{im} i$. It splits if there exists a linear section $s : C \to B$ with $p \circ s = 1_C$. In that case

$$F : A \oplus C \to B, \qquad F(a, c) = i(a) + s(c)$$

is an isomorphism preserving the inclusion and quotient maps.

Proof. If $F(a, c) = 0$, apply p to get $c = 0$, then injectivity of i gives $a = 0$. For $b \in B$, the element $b - s(p(b))$ lies in $\ker p$, so is $i(a)$ for a unique a. This also describes F^{-1}. A retraction $r : B \to A$, $ri = 1_A$, is equivalent: its kernel is a complement to $i(A)$ and restricts under p isomorphically to C. □

Over a field, lift a basis of C to B to obtain s. Over a general ring, the lifts may violate relations in C; that is the first source of extension obstructions.

The set of sections, if nonempty, is an affine space under $\operatorname{Hom}(C, A)$. For sections s, s', $p(s' - s) = 0$, so uniquely $s' - s = i \circ h$ for $h : C \to A$. Conversely $s + ih$ is a section for any h. This statement holds in module categories with the indicated Hom group. It precisely locates the ambiguity.

There is no natural complement in general. In the split model $B = A \oplus C$, the automorphism

$$u_h(a, c) = (a + h(c), c)$$

preserves i and p but sends the section $c \mapsto (0, c)$ to $c \mapsto (h(c), c)$. Whenever $h \neq 0$, no distinguished section can be invariant under all such structure-preserving automorphisms. A chosen positive inner product changes the problem: the orthogonal complement then becomes canonical relative to that extra data.

3.6 What to reconstruct and when

For an ordinary rank problem, first factor through the image. For constraints on functionals, take annihilators. For a proposed decomposition, construct its projections and check their equations. For an invariant subspace, ask explicitly whether the desired complement must preserve an operator or a group action. The underlying vector-space splitting proves less than an equivariant one.

The main source anchors are Xi II §§1.4–1.6 and 2.4 (printed pp. 14–24, 67–71; PDF pp. 20–30, 73–77), Grillet VIII.2–5 and X.1 (especially X.1.5, printed p. 395, PDF p. 407), and Mac Lane III.1–2 for the broader universal-property language. Proofs above are written out independently rather than inferred from those references.

4 Operators as modules and polynomial projections

4.1 Put the operator into the scalar ring

Let $T : V \to V$ be k-linear. Define an action of $R = k[t]$ on V by

$$p \cdot v = p(T)(v), \qquad p(T) = a_0 1_V + a_1 T + \cdots + a_d T^d \quad \text{if } p(t) = \sum_{j=0}^{d} a_j t^j.$$

Polynomial evaluation is a unital k-algebra homomorphism $k[t] \to \mathrm{End}_k(V)$, because the powers of a single operator commute. Denote the resulting module by V_T.

For another operator $S : W \to W$, a map $f : V \to W$ is $k[t]$-linear exactly when it is k-linear and

$$f \circ T = S \circ f.$$

The forward implication follows by applying f to $t \cdot v$; the reverse follows first for powers, then for polynomials. Thus operators and intertwining maps form the same category as $k[t]$-modules whose underlying k-spaces lie in the chosen dimension class. In particular, finite-dimensional operators correspond to modules finite-dimensional over k, a smaller class than all finitely generated $k[t]$-modules. Invariant subspaces are submodules, invariant quotient operators are quotient modules, and similarity means module isomorphism. This translation preserves the maps, not merely the list of isomorphism classes.

If $n = \dim_k V < \infty$, a k-basis generates V_T over R. Moreover, $1_V, T, \ldots, T^{n^2}$ are linearly dependent in $\mathrm{End}_k(V)$, so some nonzero polynomial annihilates V_T. Its annihilator ideal has a unique monic generator m_T, the minimal polynomial. For $V = 0$, set $m_T = 1$. This argument avoids using Cayley–Hamilton before proving it.

4.2 The PID structure theorem and its mechanism

Structure theorem. If R is a principal ideal domain and M is a finitely generated R-module, then

$$M \cong R^r \oplus \bigoplus_{i=1}^{s} R/(d_i), \qquad d_1 \mid d_2 \mid \cdots \mid d_s,$$

where the d_i are nonzero nonunits. The free rank and the invariant-factor ideals (d_i) are unique. Over $k[t]$, choose the d_i monic. The isomorphism and its cyclic generators are generally not canonical.

Here is the construction in the Euclidean rings relevant to ordinary linear algebra, $k[t]$ and $\mathbb{Z}$. Write M as a cokernel of a matrix $A : R^a \to R^b$. Such a finite presentation exists: a submodule of a finite free module over a PID is finite free. Invertible row and column operations change the bases of the target and source and preserve its cokernel.

If $A \neq 0$, move a nonzero entry into the first position. Divide entries in its column and row by this pivot. A nonzero remainder can be moved to the pivot and has smaller Euclidean size. Repetition ends with a pivot dividing its entire row and column, which can then be cleared. If the pivot fails to divide an entry of the remaining block, add the corresponding row to the first row; a subsequent column division produces a smaller nonzero pivot. Thus this stage also terminates. The pivot now divides the remaining block. Repeat on that block to obtain a diagonal matrix with $d_i \mid d_{i+1}$,

including units and zero entries as appropriate. Taking cokernels gives the theorem's existence statement in these Euclidean cases.

For uniqueness, let $I_j(A)$ be the ideal generated by all $j \times j$ minors. These ideals are unchanged by invertible row or column operations; multilinearity of minors gives containment in one direction and the inverse operation gives the other. For Smith diagonal form,

$$I_j(A) = (d_1 d_2 \cdots d_j)$$

for the nonzero diagonal positions, including unit positions. Thus the successive products, and hence the factors up to units, are determined. For arbitrary module presentations, invariance of the corresponding presentation ideals is a further theorem; alternatively uniqueness of the abstract decomposition follows from primary filtrations below. We are not identifying matrices of different presentation sizes without that extra argument.

For a general PID there need not be a Euclidean norm, so the specific descent algorithm above is not a proof in that generality. Grillet VIII.6.1–6.3 (printed pp. 336–342, PDF pp. 348–354) gives the general proof using principal ideals and a basis theorem. This is a cited extension of the construction just explained.

Why finite-dimensional operator modules have no free summand. The k-dimension of $k[t]$ is infinite, whereas $\dim_k V$ is finite. Hence

$$V_T \cong \bigoplus_{i=1}^{s} k[t]/(d_i), \qquad d_1 \mid \cdots \mid d_s.$$

Every block $k[t]/(d)$ has basis $1, t, \ldots, t^{d_0-1}$ if $d_0 = \deg d > 0$. Multiplication by t has companion matrix

$$C(d) = \begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{d_0-1} \end{pmatrix}, \quad d(t) = t^{d_0} + a_{d_0-1}t^{d_0-1} + \cdots + a_0.$$

The subdiagonal entries are all 1 and unspecified entries are 0; for $d_0 = 1$ this is the one-entry matrix $(-a_0)$. The defining relation $d(t) = 0$ explains the last column. Expansion of $\det(xI - C(d))$ along the last column gives $d(x)$; equivalently the cofactor of its jth last-column entry is the appropriate power of x. The invariant-factor blocks give the rational canonical form over any field. Consequently

$$\chi_T = \prod_i d_i, \qquad m_T = d_s \quad (V \neq 0).$$

In particular $m_T \mid \chi_T$, so $\chi_T(T) = 0$. This derives Cayley–Hamilton from the structure theorem, rather than assuming it in the proof that the module is torsion. For $V = 0$ the empty product is 1.

There is also a useful presentation that makes contact with matrices:

$$V_T \cong \operatorname{coker}(tI - A : R^n \to R^n),$$

where A is the matrix of T in a chosen basis. Send a vector polynomial $p(t) = \sum_j t^j v_j$ to $\sum_j A^j v_j$. The relation $tv \equiv Av$ reduces every class modulo $(tI - A)R^n$ to a constant. It also shows that a constant class maps to itself in k^n, so no nonzero constant is killed. Thus the induced map is bijective. Computing Smith form of $tI - A$ therefore computes similarity invariants of A; this is not the same as applying arbitrary left and right equivalence to A over k.

4.3 Bézout and Chinese remainder projections

Suppose $f_1, \dots, f_r \in k[t]$ are pairwise coprime and $m = f_1 \cdots f_r$ annihilates T. Set $g_i = m/f_i$. Bézout gives $a_i g_i \equiv 1 \pmod{f_i}$. Let $e_i = a_i g_i$ in $k[t]/(m)$. Its residues are 1 modulo f_i and 0 modulo every other factor. The Chinese remainder theorem, or divisibility by each coprime factor, gives

$$e_i^2 = e_i, \qquad e_i e_j = 0 \ (i \neq j), \qquad \sum_i e_i = 1 \qquad (\text{mod } m).$$

Evaluate at T to obtain projections $P_i = e_i(T)$ satisfying these same identities in $\mathrm{End}_k(V)$. Their images are exactly $\ker f_i(T)$: first $f_i e_i$ is divisible by m, so the image lies there; conversely $e_i - 1$ is divisible by f_i, so $e_i(T)$ fixes that kernel. Hence

$$V = \bigoplus_i \ker f_i(T).$$

The polynomials a_i are not unique, but the projections are uniquely determined by the specified factors and the action. If m is the minimal polynomial and the factors are its primary powers, these are the canonical primary summands. Any operator commuting with T commutes with all the P_i, and thus preserves these summands.

Small example. If $T^2(T - 1_V) = 0$, take $P_1 = T^2$ and $P_0 = 1_V - T^2$. Indeed $T^3 = T^2$ implies $T^4 = T^2$, so $P_1^2 = P_1$, and

$$V = \ker T^2 \oplus \ker(T - 1_V).$$

With $T^2(T - 1_V)^2 = 0$ instead, one may take

$$P_1 = 3T^2 - 2T^3, \qquad P_0 = 1_V - 3T^2 + 2T^3.$$

These integer-coefficient formulas work in every characteristic: their defining congruences are polynomial identities over $\mathbb{Z}$. There is no division by 2 or 3. Only the distinctness of 0 and 1 is used.

What fails without coprimality. Take $R/(t^2)$. Its submodule $tR/(t^2)$ and quotient $R/(t)$ are both one-dimensional over k. They do not give an invariant direct sum: the only vectors killed by t are in $tR/(t^2)$. Dividing one repeated primary factor into smaller pieces does not create Bézout projectors.

4.4 Jordan form and what is actually canonical

Factor the invariant factors into monic irreducibles. Chinese remaindering gives blocks $k[t]/(p^a)$. This elementary-divisor decomposition exists over every field. If the minimal polynomial splits over

k, each $p = t - \lambda$. Writing $N = T - \lambda 1_V$ on a block of size a, choose a cyclic vector v with $N^{a-1}v \neq 0$ and $N^a v = 0$. The ordered basis

$$N^{a-1}v, N^{a-2}v, \dots, Nv, v$$

gives the Jordan block $J_a(\lambda)$ with 1 on the superdiagonal. Thus Jordan form requires splitting of m_T over k, not necessarily that k itself be algebraically closed.

To recover block sizes without choosing chains, put $a_j = \dim \ker (T - \lambda 1_V)^j$ and $a_0 = 0$. Every λ-block of size b contributes $\min(j, b)$; other primary blocks contribute zero because $T - \lambda 1_V$ is invertible there. Therefore

$$\begin{aligned} a_j - a_{j-1} &= \#\{\lambda\text{-blocks of size at least } j\}, \\ 2a_j - a_{j-1} - a_{j+1} &= \#\{\lambda\text{-blocks of size exactly } j\}. \end{aligned}$$

For example, nullities $0, 3, 5, 6, 6$ give three blocks, two of size at least 2, one of size at least 3, and none of size at least 4: the sizes are $3, 2, 1$. This gives uniqueness and a practical reconstruction procedure. More generally a p^b-block contributes $\deg(p) \min(j, b)$ to $\dim_k \ker p(T)^j$, so these filtrations recover the elementary divisors over a nonsplitting field as well.

The primary subspaces and block-size multiset are intrinsic. Individual cyclic summands, Jordan chains, their ordering, and a conjugating matrix are choices. Conflating a canonical list of invariants with a canonical basis leads to many false naturality claims.

Diagonalizability and semisimplicity. T is diagonalizable over k if and only if m_T is a product of distinct linear factors. The $k[t]$-module V_T is semisimple if and only if m_T is a product of distinct monic irreducible factors. These differ when the field does not split m_T. A real plane rotation through a nonzero angle other than π can be a simple $\mathbb{R}[t]$-module without having a real eigenvector.

4.5 Hom spaces and the commutant

A module map $R/(f) \to R/(g)$ is determined by the image y of 1, and its well-definedness condition is $fy = 0$ in $R/(g)$. For monic $d = \gcd(f, g)$ write $f = df_0$, $g = dg_0$ with $\gcd(f_0, g_0) = 1$. The condition $g \mid fy$ is equivalent to $g_0 \mid y$. Thus

$$\operatorname{Hom}_R(R/(f), R/(g)) \cong R/(d), \qquad h \bmod d \longmapsto (1 \longmapsto g_0 h \bmod g).$$

Cancellation here takes place in the domain $k[t]$. In particular

$$\dim_k \operatorname{Hom}_R(R/(f), R/(g)) = \deg \gcd(f, g).$$

The commutant of T is $\operatorname{End}_{k[t]}(V_T)$. If the elementary divisors are $(t - \lambda)^{b_{\lambda,i}}$, then

$$\dim_k \{S : ST = TS\} = \sum_{\lambda} \sum_{i,j} \min(b_{\lambda,i}, b_{\lambda,j}).$$

This computes the number of independent commuting maps without solving all matrix-entry equations. For a single cyclic block its commutant is $k[T]$. For multiple identical blocks there are extra intertwiners, so a commuting operator need not be a polynomial in T.

4.6 Fitting decomposition and the boundary of the method

For $N \geq \dim V$, both the kernels and images of powers of T have stabilized. Then

$$V = \ker T^N \oplus \operatorname{im} T^N.$$

If $v = T^N w$ lies in the intersection, then $T^{2N} w = 0$; stabilization gives $w \in \ker T^N$, so $v = 0$. Rank–nullity gives the direct sum. The restriction to the first summand is nilpotent and to the second is bijective. These subspaces are canonical, though a basis for either is not. This is the primary split into the factor t and the factors coprime to it, obtained without explicitly factoring the minimal polynomial.

For two commuting operators the ring is $k[s,t]$. It is not a PID: the ideal (s,t) is not principal. A simultaneous eigenbasis exists for a finite family of commuting diagonalizable operators over a common splitting field: decompose into eigenspaces of one operator and note that the others preserve them. Their restrictions remain diagonalizable because their minimal polynomials divide the original square-free split minimal polynomials, so induction applies. Without diagonalizability, the one-variable module classification does not provide simultaneous Jordan form.

Source route. Xi II Theorems 2.51–2.57 and the existence/uniqueness proofs (printed pp. 72–77; PDF pp. 78–83) already contain Bézout decomposition, quotient induction, and intrinsic block counts. Grillet VIII.6–7 supplies the module derivation (printed pp. 336–346; PDF pp. 348–358); Hungerford IV.6 and VII.4–5 supplies another module-first route. The presentation, centralizer, and Fitting arguments above are included so the route is operational rather than a list of references.

5 Hom and tensor products

5.1 Hom records maps and variance

$\operatorname{Hom}_k(V,W)$ is a vector space under pointwise operations. Maps $a : V' \to V$ and $b : W \to W'$ induce

$$\operatorname{Hom}(V,W) \to \operatorname{Hom}(V',W'), \qquad f \longmapsto b \circ f \circ a.$$

The source variable reverses arrows and the target variable preserves them. For modules over a noncommutative ring, $\operatorname{Hom}_R(M,N)$ is always an abelian group but generally has no natural left R-module structure. Commutativity or specified bimodule actions must be supplied before treating it as one.

The kernel universal property gives left exactness. For a short exact sequence $0 \to A \xrightarrow{i} B \xrightarrow{p} C \to 0$ of R-modules,

$$0 \to \operatorname{Hom}_R(X,A) \xrightarrow{i\circ -} \operatorname{Hom}_R(X,B) \xrightarrow{p\circ -} \operatorname{Hom}_R(X,C)$$

is exact. A map $X \to B$ killed by p factors uniquely through i. There need not be surjectivity at the right: lifting the identity of C would be a section of p. Dually,

$$0 \to \operatorname{Hom}_R(C,X) \to \operatorname{Hom}_R(B,X) \to \operatorname{Hom}_R(A,X)$$

is exact. A map on the submodule A need not extend to B. Over a field both failures disappear, by choosing a splitting. The resulting exactness is canonical; its proof need not make a canonical choice.

5.2 Tensor universality and a construction

A tensor product of vector spaces V, W is a vector space $V \otimes_k W$ together with a bilinear map $\tau : V \times W \to V \otimes_k W$ such that every bilinear map $\beta : V \times W \to X$ factors uniquely as

$$\beta = \tilde{\beta} \circ \tau, \qquad \tilde{\beta} : V \otimes_k W \to X \text{ linear.}$$

Write $v \otimes w = \tau(v, w)$. The pair, not the bare vector space, is the universal object.

For existence, take the free vector space on symbols $[v, w]$ and quotient by the span of all bilinearity relations:

$$[v + v', w] - [v, w] - [v', w], \quad [v, w + w'] - [v, w] - [v, w'],$$
$$[av, w] - a[v, w], \quad [v, aw] - a[v, w].$$

Every bilinear β extends linearly from the symbols and annihilates these relations, hence factors through the quotient. The symbols span, giving uniqueness. Universal uniqueness is proved by composing the two comparison maps, as for quotients.

If (e_i) and (f_j) are bases, then $(e_i \otimes f_j)$ is a basis. Spanning follows by expansion. For independence, the coefficient maps $\beta_{ij}(v, w) = e^i(v) f^j(w)$ induce linear functionals on the tensor product extracting each coefficient of a finite relation. This proof works for infinite algebraic bases too, because each vector and tensor uses only finite sums. It does not assert that dual bases span the whole algebraic dual in that case.

Recognition signal. A bilinear expression should usually be turned into a linear map on a tensor product before choosing coordinates. A formula on pure tensors defines a linear map only after its bilinearity or balancing has been checked. Pure tensors span; they are not generally all the elements.

For instance, $e_1 \otimes f_1 + e_2 \otimes f_2$ is not pure when the two displayed pairs are independent. Under coefficient matrices it has rank 2, whereas a pure tensor has entries $v_i w_j$ and rank at most 1.

5.3 Functorial maps, symmetry, and associativity

For $a : V \to V'$ and $b : W \to W'$, bilinearity gives the unique linear map

$$a \otimes b : V \otimes W \to V' \otimes W', \qquad (a \otimes b)(v \otimes w) = a(v) \otimes b(w).$$

Checking on pure tensors proves $(a' \otimes b') \circ (a \otimes b) = (a' \circ a) \otimes (b' \circ b)$. The same method gives canonical isomorphisms

$$V \otimes W \cong W \otimes V, \qquad (U \otimes V) \otimes W \cong U \otimes (V \otimes W),$$
$$k \otimes V \cong V, \qquad (U \oplus V) \otimes W \cong (U \otimes W) \oplus (V \otimes W).$$

These identify objects by specified maps; they are not literal equalities of set-theoretic constructions. Associativity is especially transparent by observing that both iterated tensor products represent trilinear maps.

In chosen product bases, the matrix of $a \otimes b$ is the Kronecker product. If A is $n \times n$ and B is $m \times m$,

$$\operatorname{tr}(A \otimes B) = \operatorname{tr}(A)\operatorname{tr}(B), \quad \det(A \otimes B) = \det(A)^m \det(B)^n.$$

For trace, sum the diagonal entries $a_{ii}b_{jj}$. For determinant, write $A \otimes B = (A \otimes I_m)(I_n \otimes B)$; the second factor is a direct sum of n copies of B, and the first is conjugate by the basis permutation $e_i \otimes f_j \mapsto f_j \otimes e_i$ to a direct sum of m copies of A. This proves the identity over every field and also over commutative rings. No diagonalization hypothesis is required.

5.4 Currying and the Hom–tensor adjunction

There is a canonical isomorphism, with no finite-dimensional hypothesis,

$$\operatorname{Hom}_k(U \otimes V, W) \cong \operatorname{Hom}_k(U, \operatorname{Hom}_k(V, W)).$$

It sends F to the map $u \mapsto (v \mapsto F(u \otimes v))$. Conversely a linear $g : U \to \operatorname{Hom}(V, W)$ yields the bilinear map $(u, v) \mapsto g(u)(v)$, hence a unique F. These operations are inverse, and pre/postcomposition proves naturality in all variables.

One can remember this as: "a linear map out of a tensor is a bilinear map, and a bilinear map is a linear family of linear maps." The adjunction is the precise version of that sentence, not an extra theorem to memorize separately.

For modules over a commutative ring R the same formulas work. Over a noncommutative ring one uses a right module and a left module and tracks the remaining bimodule actions. A correct underlying abelian-group version is

$$\operatorname{Hom}_{\mathbb{Z}}(M \otimes_R N, A) \cong \operatorname{Hom}_{R^{\mathrm{op}}}(M, \operatorname{Hom}_{\mathbb{Z}}(N, A)),$$

where M is right R, N is left R, and $(\phi \cdot r)(n) = \phi(rn)$ gives the Hom group its right action. The tensor represents biadditive maps satisfying $\beta(mr, n) = \beta(m, rn)$, called balanced maps. Two arbitrary left modules over a noncommutative ring cannot simply be tensored over R using the commutative formulas.

5.5 Finite-dimensional Hom as a tensor and trace as evaluation

Define

$$\Theta : W \otimes V^* \to \operatorname{Hom}(V, W), \qquad \Theta(w \otimes \lambda)(v) = \lambda(v)w.$$

If V is finite-dimensional, its inverse sends f to $\sum_i f(e_i) \otimes e^i$. Indeed evaluation at $v = \sum_i e^i(v)e_i$ recovers $f(v)$. Conversely applying the inverse to a pure tensor recovers it. The inverse expression is basis-independent because it is the inverse of a specified canonical map. W need not be finite-dimensional for this result.

For $W = V$, compose Θ^{-1} with the evaluation contraction $v \otimes \lambda \mapsto \lambda(v)$. This defines trace invariantly, and the basis formula is $\operatorname{tr} T = \sum_i e^i(T(e_i))$. The tensor corresponding to 1_V is $\sum_i e_i \otimes e^i$; it is basis-independent even though each summand is not.

Composition of rank-one maps is

$$\Theta(v \otimes \lambda) \circ \Theta(w \otimes \mu) = \lambda(w)\Theta(v \otimes \mu).$$

Taking traces gives $\lambda(w)\mu(v)$ in either order. Rank-one maps span all endomorphisms, so $\operatorname{tr}(ST) = \operatorname{tr}(TS)$. This is an evaluation identity, which explains why trace is invariant under similarity and why every commutator has trace zero. The converse is not needed here, and in positive characteristic trace of the identity can be zero without the identity being zero.

For infinite-dimensional V, every tensor in $W \otimes V^*$ still maps to a finite-rank operator, and every finite-rank operator arises this way. The canonical map is injective: express a tensor with linearly independent W-components and use coordinate functionals to show its coefficients vanish if the operator vanishes. Thus its image is exactly the finite-rank maps. When $V = W$ is infinite-dimensional the identity is a concrete obstruction to surjectivity.

This also exposes the issue in Xi II's remark after Theorem 6.19, printed p. 278 (PDF p. 284). The preceding definition uses **all** multilinear forms on $V^p \times (V^*)^q$. For $p = q = 1$, evaluation $(v, \lambda) \mapsto \lambda(v)$ is such a form, but it cannot be a finite sum $\sum_i \alpha_i(v)\lambda(w_i)$: that would represent 1_V as a finite-rank map. The finite-dimensional theorem is sound; its unqualified infinite-dimensional extension needs to be withdrawn or the tensor space restricted. The printed remark and definition were both checked as page images, so this is a mathematical correction, not an OCR inference.

5.6 Exactness, quotients, and change of scalars

Over a field, tensoring is exact. One proof chooses a splitting of a short exact sequence and uses distribution over direct sums. Alternatively a basis of W identifies $- \otimes W$ with a direct sum of copies of the original map. These are proofs of a canonical exactness statement, not canonical splittings.

Over a commutative ring, tensoring is always right exact:

$$A \xrightarrow{f} B \longrightarrow C \longrightarrow 0 \quad \Longrightarrow \quad A \otimes N \xrightarrow{f \otimes 1} B \otimes N \longrightarrow C \otimes N \longrightarrow 0.$$

To see the kernel, take the quotient of $B \otimes N$ by the image of $A \otimes N$. Bilinear maps from $B \times N$ vanishing on that image are precisely the ones factoring through $C \times N$. The quotient and $C \otimes N$ therefore satisfy the same universal property. In particular

$$(M/U) \otimes_R N \cong (M \otimes_R N)/\operatorname{im}(U \otimes_R N).$$

The word "image" matters: the induced map $U \otimes_R N \to M \otimes_R N$ need not be injective. For $0 \to \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \to \mathbb{Z}/2 \to 0$, tensoring with $\mathbb{Z}/2$ turns the first displayed map into zero. Thus a tensor product does not always preserve subobjects. A module for which tensoring preserves all short exact sequences is called flat.

If L/k is a field extension, $L \otimes_k V$ has an L-action $a(b \otimes v) = ab \otimes v$. A k-basis (e_i) gives an L-basis $(1 \otimes e_i)$, so dimension over the **new** field remains $\dim_k V$. The map $v \mapsto 1 \otimes v$ is injective, and scalar extension is exact. A real operator may acquire eigenvectors after extending to $\mathbb{C}$; its complexified invariant decomposition need not descend to real one-dimensional subspaces. The coefficient field is part of every classification statement.

Sources. Xi II §6.2, especially Definition 6.11, Theorem 6.15, and Examples 6.13–6.14 (printed pp. 273–278; PDF pp. 279–284), contains actual tensor universality and its applications. Grillet XI.5–8 (printed pp. 434–456; PDF pp. 446–468) supplies balanced products, general exactness, dual modules, and flatness. The concrete calculations here show why that extra generality matters.

6 Free algebras, exterior powers, determinants, and minors

6.1 Free constructions remember their extension property

The tensor algebra of a vector space is

$$T(V) = \bigoplus_{r \geq 0} V^{\otimes r}, \qquad V^{\otimes 0} = k,$$

with multiplication by concatenation of tensors. Each element has finitely many nonzero degree components. The inclusions of k and V are part of the construction. For any unital associative k-algebra A, a linear map $f : V \to A$ extends uniquely to an algebra homomorphism

$$\hat{f} : T(V) \to A, \qquad \hat{f}(v_1 \otimes \cdots \otimes v_r) = f(v_1) \cdots f(v_r).$$

The product on the right is multilinear, giving a map on each tensor power; concatenation proves multiplicativity, and generation by V proves uniqueness. An algebra generated by a specified copy or image of V is therefore a quotient of $T(V)$ by its relations. The quotient has to impose exactly the intended identities; it cannot be guessed from a superficial resemblance of formulas.

The symmetric algebra is

$$\mathrm{Sym}(V) = T(V)/\langle v \otimes w - w \otimes v : v, w \in V \rangle.$$

It is the free **commutative** unital algebra on V. The extension above factors precisely when the images $f(v)$ commute. With a basis $e_1, \ldots, e_n$, sending e_i to a variable x_i identifies it with $k[x_1, \ldots, x_n]$. The variables and monomial basis depend on the chosen basis, while the free property does not.

The homogeneous degree-r part $\mathrm{Sym}^r(V)$ represents symmetric r-linear maps from V^r. Notice the variance: polynomial expressions on a space V belong to $\mathrm{Sym}(V^*)$, not $\mathrm{Sym}(V)$, once its vectors are regarded as points. Over a finite field, formal polynomials are not determined by their values on all points: $x^q - x$ vanishes on $\mathbb{F}_q$ but is a nonzero polynomial. Thus a coordinate ring should not silently be replaced by the ring of set functions on its rational points.

6.2 Exterior algebra: alternation is the relation

Define

$$\Lambda(V) = T(V)/\langle v \otimes v : v \in V \rangle.$$

The product is denoted $\wedge$. Expanding $(v + w) \wedge (v + w) = 0$ gives $v \wedge w = -w \wedge v$, and exchanging blocks gives $\alpha \wedge \beta = (-1)^{rs} \beta \wedge \alpha$ for homogeneous degrees r, s. The defining equation $v \wedge v = 0$ remains necessary in characteristic 2: a sign rule alone does not imply it there.

The degree-r piece $\Lambda^r(V)$ represents alternating r-linear maps: an alternating map vanishes whenever two arguments are equal. Its unique factorization through $V^{\otimes r}$ kills the repeated-vector relations; conversely the wedge map is alternating. This construction works over any commutative ring, not only in characteristic zero.

With an ordered basis $e_1, \dots, e_n$, the wedges

$$e_{i_1} \wedge \cdots \wedge e_{i_r}, \qquad 1 \le i_1 < \cdots < i_r \le n$$

form a basis. Reordering and removing repetitions proves spanning. For independence, for each ordered subset $I = (i_1, \dots, i_r)$ define on vectors with coordinates v_{ab} the alternating coefficient form

$$\beta_I(v_1, \dots, v_r) = \sum_{\sigma \in S_r} \operatorname{sgn}(\sigma) \prod_{a=1}^{r} v_{i_a, \sigma(a)}.$$

When two columns coincide, pair the permutations exchanging those columns; their terms sum to zero, in characteristic 2 as well. Thus β_I factors through $\Lambda^r(V)$ and takes value 1 on the I basis wedge and 0 on the other ascending ones. This proves independence without dividing by $r!$. Therefore

$$\dim \Lambda^r(V) = \binom{n}{r}, \qquad \Lambda^r(V) = 0 \ (r > n).$$

For $r = 0$, $\Lambda^0(V) = k$. The zero-dimensional top exterior power is therefore a line too.

6.3 Antisymmetrizing tensors is a model with extra hypotheses

If $r!$ is invertible in k, the operator

$$\operatorname{Alt}_r = \frac{1}{r!} \sum_{\sigma \in S_r} \operatorname{sgn}(\sigma)\sigma : V^{\otimes r} \to V^{\otimes r}$$

is idempotent. Its image identifies with $\Lambda^r(V)$: the quotient map restricted to that image is inverse to the map induced by Alt_r. With this normalization, exterior multiplication in the image is $\alpha \wedge \beta = \operatorname{Alt}_{r+s}(\alpha \otimes \beta)$. Other common conventions place factorial factors in the identification, so one must check which model a formula uses.

The analogous averaging operator without signs models symmetric powers when $r!$ is invertible. In small characteristic the quotient definitions remain valid but these normalized projectors do not. Symmetric powers (coinvariants) and invariant symmetric tensors must not be identified using a nonexistent $1/r!$. Polarization formulas with factorial denominators have the same limitation. Xi II §6.4 explicitly assumes characteristic zero for its antisymmetric-tensor model (printed p. 292, PDF p. 298); Grillet XIII.4 uses the quotient construction over a commutative ring (printed pp. 523–527, PDF pp. 535–539). These are related constructions with different stated scope.

6.4 Functoriality, rank, and determinants

For $f : V \to W$, the alternating map $(v_1, \dots, v_r) \mapsto f(v_1) \wedge \cdots \wedge f(v_r)$ induces

$$\Lambda^r f : \Lambda^r V \to \Lambda^r W.$$

Checking decomposable wedges gives $\Lambda^r(g \circ f) = \Lambda^r g \circ \Lambda^r f$ and $\Lambda^r 1_V = 1_{\Lambda^r V}$. For $r > 0$,

$$\Lambda^r f \neq 0 \quad \iff \quad \operatorname{rank} f \geq r.$$

If the image has dimension less than r, every image wedge is zero. If the rank is at least r, choose r vectors whose images are independent and extend those images to a basis of W; their wedge is nonzero by the basis theorem. This is the intrinsic meaning of testing rank by minors.

For $n = \dim V$ define the determinant line $\det V = \Lambda^n V$. It is one-dimensional. An endomorphism $T : V \to V$ therefore acts on it by a unique scalar:

$$\Lambda^n T = (\det T) 1_{\det V}.$$

This scalar is the determinant. Functoriality immediately gives $\det(ST) = \det S \det T$, and $\det T \neq 0$ precisely when T is invertible. For $V = 0$ the determinant is 1 by the degree-zero convention. Expanding $T(e_1) \wedge \cdots \wedge T(e_n)$ gives the usual signed permutation formula. Thus elimination formulas can be reconstructed from alternation and the chosen volume basis.

If $f : V \to W$ maps two different n-dimensional spaces, it determines a map $\det V \to \det W$. There is no preferred scalar determinant until trivializations of both lines are chosen. This distinction is useful in coordinate changes, orientation, and families of vector spaces.

6.5 Minors and Cauchy–Binet from composition

Let A be the matrix of $f : k^n \to k^m$. In ascending wedge bases, the entry of $\Lambda^r f$ in row subset I and column subset J is the minor $\det A_{I,J}$. This follows by expanding the image of $e_{j_1} \wedge \cdots \wedge e_{j_r}$ and collecting the I coefficient.

For a second matrix $B : k^m \to k^\ell$, functoriality gives

$$\det(BA)_{I,J} = \sum_{\substack{K \subseteq \{1,\dots,m\} \\ |K| = r}} \det B_{I,K} \det A_{K,J}.$$

This is the full minors version of Cauchy–Binet. The familiar determinant formula for a rectangular product is the case $r = n = \ell$. The same proof works over commutative rings using finite free modules and the exterior basis theorem. A rank interpretation over a field should not be carried unchanged to rings with zero divisors.

Expanding $\det(xI - T)$ by the columns in which $-T$ is selected gives

$$\chi_T(x) = \sum_{r=0}^{n} (-1)^r \operatorname{tr}(\Lambda^r T) x^{n-r}.$$

The trace of $\Lambda^r T$ is the sum of principal r-minors. In particular $\operatorname{tr} T$ and $\det T$ are the first and last nontrivial elementary symmetric coefficients. This works without eigenvalues in k. In positive characteristic, recovering these coefficients from power traces via Newton identities can require division by a nonunit integer; the exterior-power formula itself does not.

6.6 Determinant lines of exact sequences

For a short exact sequence of finite-dimensional vector spaces $0 \to A \xrightarrow{i} B \xrightarrow{p} C \to 0$, put $a = \dim A$ and $c = \dim C$. There is a canonical isomorphism

$$\det A \otimes \det C \xrightarrow{\sim} \det B.$$

Send a decomposable tensor to

$$(a_1 \wedge \cdots \wedge a_a) \otimes (c_1 \wedge \cdots \wedge c_c) \longmapsto i(a_1) \wedge \cdots \wedge i(a_a) \wedge \tilde{c}_1 \wedge \cdots \wedge \tilde{c}_c,$$

where $p(\tilde{c}_j) = c_j$. Changing a lift adds a vector of $i(A)$, whose extra occurrence makes that term zero. Multilinearity and alternation give a map on the determinant lines, and lifting a basis of C gives a basis of B, so it is an isomorphism. It is independent of the chosen lifts, although the splitting of the original sequence is not. The order "A then C" is part of the convention; reversing it contributes $(-1)^{ac}$.

An operator preserving A therefore satisfies $\det T_B = \det T_A \det T_C$. Applied to $xI - T$ over $k(x)$, or directly to its block triangular matrix, this gives $\chi_{T_B} = \chi_{T_A}\chi_{T_C}$ even without an invariant complement. Multiplication of invariants across an exact sequence does not imply splitting.

6.7 Contraction and reconstruction

For $\lambda \in V^*$ define contraction $\iota_\lambda : \Lambda^r V \to \Lambda^{r-1} V$ by

$$\iota_\lambda(v_1 \wedge \cdots \wedge v_r) = \sum_{j=1}^{r} (-1)^{j-1} \lambda(v_j) v_1 \wedge \cdots \widehat{v}_j \cdots \wedge v_r.$$

The right side is alternating and multilinear in the v_j, so the universal property makes it well defined. Splitting the sum into the positions of α and β gives

$$\iota_\lambda(\alpha \wedge \beta) = (\iota_\lambda \alpha) \wedge \beta + (-1)^r \alpha \wedge \iota_\lambda \beta \quad (\deg \alpha = r).$$

This supplies a controlled introduction to signs in graded constructions. Exterior powers, contractions, and determinant lines are useful now; differential forms additionally require a manifold and smooth variation. They should not be identified merely because the same wedge symbol appears.

The core memory is short: free algebra extends linear maps, quotient imposes relations, exterior power transports alternating data, and the top exterior power is a line. Minors, rank tests, determinant multiplication, and Cauchy–Binet can be rebuilt from those principles.

7 Forms, adjoints, and the correct equivalence relation

7.1 Similarity, equivalence, and congruence classify different data

For an endomorphism $T : V \to V$, changing the basis by columns S changes its matrix from A to $S^{-1}AS$. The source and target are the same space and use the same changed basis: this is similarity.

For a map $f : V \to W$ with independently changed source and target bases, the matrix becomes $Q^{-1}AP$. This is matrix equivalence; over a field its only invariant is rank.

For a bilinear form $b : V \times V \to k$ with matrix B, $b(x, y) = x^{\mathsf{T}}By$. The same change of vector basis in both arguments gives $B \mapsto S^{\mathsf{T}}BS$, called congruence. It is not similarity. The associated map

$$b^{\flat} : V \to V^*, \qquad b^{\flat}(v)(w) = b(v, w)$$

has a dual space as target; the two basis changes are linked by duality. A nondegenerate form is exactly one for which this map is an isomorphism. This identification depends on b.

For a Hermitian form over $\mathbb{C}$, this study uses the convention

$$h(x, y) = \overline{x}^{\mathsf{T}}Hy,$$

conjugate-linear in the first argument and linear in the second, with $H = \overline{H}^{\mathsf{T}}$. The matrix changes by $H \mapsto S^*HS$ where $S^* = \overline{S}^{\mathsf{T}}$. The map $v \mapsto h(v, -)$ into the algebraic dual is conjugate-linear. Xi's inner-product convention is linear in the first argument instead; translate the convention rather than copying an adjoint formula unchanged.

A decisive example. Over $\mathbb{R}$, I_2 and $2I_2$ are congruent via $S = \sqrt{2}I_2$, but are not similar because they have different eigenvalues. Conversely the real symmetric matrices $\operatorname{diag}(1, -1)$ and $\operatorname{diag}(2, -2)$ have the same inertia but different similarity classes. A theorem about Jordan form cannot replace a theorem about forms.

7.2 Symmetric forms and quadratic forms

Assume $\operatorname{char} k \neq 2$. A symmetric bilinear form determines $q(v) = b(v, v)$, and

$$b(v, w) = \frac{q(v + w) - q(v) - q(w)}{2}.$$

Thus symmetric forms and quadratic forms encode the same data in this characteristic. In characteristic 2, $b(v, v)$ can be identically zero for a nonzero alternating form, and a quadratic form needs its own definition and polarization theory. Dividing by 2 conceals an essential hypothesis.

Every symmetric bilinear form over a field of characteristic different from 2 has a diagonal matrix in some basis. If $b \neq 0$, polarization shows there is v with $b(v, v) \neq 0$. Then for any w,

$$w = \frac{b(v, w)}{b(v, v)}v + \left(w - \frac{b(v, w)}{b(v, v)}v\right)$$

is a decomposition into kv and its b-orthogonal complement. The sum is direct since $b(v, v) \neq 0$. Induct on dimension. If the remaining restriction is zero, it contributes a radical summand. This

proof explains both diagonalization and its field dependence: nonzero diagonal entries can be rescaled only by squares available in k.

Over $\mathbb{R}$, rescale the nonzero coefficients to $+1$ or -1. The numbers (p, q, r) of positive, negative, and zero coefficients are unique (Sylvester's law of inertia). To see uniqueness of p, the diagonal form has a positive-definite subspace of dimension p. Any subspace of dimension greater than p meets the span of the negative and zero coordinate vectors nontrivially by the dimension inequality, and hence cannot be positive definite. Thus p is the maximal dimension of a positive-definite subspace, an intrinsic number. The same argument gives q, and r is the dimension of the radical.

Over an algebraically closed field of characteristic different from 2, every nonzero coefficient is a square, so rank classifies symmetric forms up to congruence. Over $\mathbb{Q}$ or a number field, rank alone does not: square classes and further arithmetic invariants matter. "Diagonal" is not the same as "classified."

7.3 Alternating forms and symplectic pairs

A bilinear form is alternating if $b(v, v) = 0$ for all v. This implies $b(v, w) = -b(w, v)$ in every characteristic. Conversely skew-symmetry implies alternation only when $2 \neq 0$.

If $b \neq 0$, choose u, v with $b(u, v) \neq 0$ and rescale to $b(u, v) = 1$. The span $P = \langle u, v \rangle$ has matrix $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and is nondegenerate. For any w define

$$w' = w + b(v, w)u - b(u, w)v.$$

Then $b(u, w') = b(v, w') = 0$, so $V = P \oplus P^{\perp}$. Induction decomposes an alternating form into symplectic pairs and a zero radical. Its rank is even, and a nondegenerate alternating space has even dimension. This argument works in characteristic 2 with the definition of alternation retained.

The essential memory is the nondegenerate two-dimensional pair and the orthogonal-complement induction, not a special sequence of row operations.

7.4 Adjoint operators require a chosen form

Let h be a nondegenerate Hermitian form on a finite-dimensional complex space. There is a unique $T^{\dagger}$ such that

$$h(Tv, w) = h(v, T^{\dagger}w) \qquad (v, w \in V).$$

In a basis with Gram matrix H, its matrix is

$$[T^{\dagger}] = H^{-1}[T]^{*}H.$$

This proves existence and uniqueness and shows explicitly why the adjoint is ordinary conjugate transpose only in an orthonormal basis. It also gives

$$(ST)^{\dagger} = T^{\dagger}S^{\dagger}, \qquad (aS + bT)^{\dagger} = \bar{a}S^{\dagger} + \bar{b}T^{\dagger},$$
$$(T^{\dagger})^{\dagger} = T.$$

For a nondegenerate real symmetric form the analogous formula is $H^{-1}A^{\mathsf{T}}H$. The algebraic dual map T^* is defined without a form and has a different type; the adjoint uses the form to return to V.

If h is positive definite, every subspace has a canonical orthogonal complement relative to h. Moreover

$$U \text{ is } T\text{-invariant} \quad \Longleftrightarrow \quad U^{\perp} \text{ is } T^{\dagger}\text{-invariant.}$$

For example, if $u \in U$ and $w \in U^{\perp}$, $h(u, T^{\dagger}w) = h(Tu, w) = 0$. This is the mechanism behind invariant orthogonal decompositions. For an indefinite nondegenerate form a subspace can intersect its own orthogonal complement: the isotropic line $\mathbb{R}(1,1)$ for $\operatorname{diag}(1,-1)$ does. Nondegeneracy of the ambient form alone does not give a direct sum with every orthogonal complement.

7.5 The finite-dimensional spectral theorem, with the hypotheses visible

Theorem. On a finite-dimensional positive-definite complex inner-product space, T has an orthonormal eigenbasis if and only if $TT^{\dagger} = T^{\dagger}T$.

Proof. One direction follows from a diagonal matrix and its conjugate transpose. For the other, call T normal. Any $N = T - \lambda 1_V$ is normal and

$$\|Nv\|^2 = h(v, N^{\dagger}Nv) = h(v, NN^{\dagger}v) = \|N^{\dagger}v\|^2.$$

Hence $Tv = \lambda v$ implies $T^{\dagger}v = \bar{\lambda}v$. The complex characteristic polynomial has a root, so choose a unit eigenvector v. The line kv is invariant under both T and $T^{\dagger}$; its orthogonal complement is therefore also invariant under both. The restrictions commute and are adjoints there, so induction gives the orthonormal eigenbasis. □

A self-adjoint operator has real eigenvalues: if $Tv = \lambda v$, then $h(v, Tv) = \lambda\|v\|^2$ is real. A unitary operator has eigenvalues of modulus 1 by norm preservation. Real self-adjoint operators admit real orthonormal eigenbases, for instance by complexifying and observing that real eigenvalues yield real eigenspaces of the required dimensions. A real normal operator need not have real eigenvectors: a planar rotation gives a 2×2 real block instead. The field and the positive metric are doing actual work.

The nonzero nilpotent matrix $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has a Jordan basis, but no orthonormal eigenbasis. It is not normal under the standard positive inner product. A purely module-theoretic classification does not replace the spectral theorem's metric conclusion.

7.6 Gram matrices, least squares, and determinant bounds

For vectors $v_1, \dots, v_r$ in a positive inner-product space, their Gram matrix is $G = (h(v_i, v_j))$. If X is the matrix with these vectors as columns in an orthonormal basis, then $G = X^*X$. Thus

$$c^*Gc = \left\| \sum_j c_j v_j \right\|^2 \geq 0,$$

and G is positive definite exactly when the vectors are independent. This is the correct reason the normal-equations matrix X^*X is invertible for a full-column-rank least-squares problem. If X is rank

deficient, the minimizer in coefficient space need not be unique, though the orthogonal projection of the target onto $\operatorname{im} X$ is unique.

Gram–Schmidt writes independent vectors as $X = QR$ with orthonormal columns of Q and upper triangular R. Its diagonal entries are the lengths of successive orthogonal residuals, so

$$\det G = \prod_j |r_{jj}|^2 \le \prod_j \|v_j\|^2.$$

Dependent vectors give determinant zero, so the inequality holds in general. Equality for nonzero vectors occurs precisely when all projections onto earlier vectors vanish, equivalently the vectors are pairwise orthogonal. For a square real matrix this yields Hadamard's determinant inequality $|\det X| \le \prod_j \|v_j\|$. A sign matrix with orthogonal rows attains the bound, linking metric geometry directly to the intended Hadamard problem without using an unnecessary abstraction.

7.7 Affine and projective information that must be retained

For an affine quadratic polynomial $q(x) = b(x, x) + \ell(x) + c$ in characteristic different from 2, a translation $x = y + a$ changes its linear part to $\ell + 2b(a, -)$. If b is nondegenerate one can choose a unique a to cancel that part. If b is degenerate, the component of ℓ on the radical cannot be removed this way and controls further geometric types. The constant and the permitted transformations also matter. Similarity invariants of a coefficient matrix do not classify affine quadrics.

Projectivizing identifies nonzero scalar multiples of a vector, and homogeneous quadratic equations are themselves defined up to nonzero scale. Real points, signature, isotropic directions, and the distinction between affine and projective transformations are information a general algebra text may not teach in the same way as Xi II Chapters 4–5. Those chapters are selectively deferable for the operator route; their geometric content is not thereby proved replaceable.

Source anchors. Xi II §1.7 (printed pp. 25–43), Chapter 3 (pp. 85–160), and Chapters 4–5 (pp. 161–262) supply forms, adjoints, metric examples, orthogonal projection, and geometry. The normal spectral theorem is Theorem 3.62 (printed pp. 126–128; PDF pp. 132–134). The book comparison distinguishes these substantial results from the graduate books' module and category-theoretic strengths.

8 Exact sequences, extensions, and calibrated homological algebra

8.1 Exactness records how pieces fit

A sequence $A \xrightarrow{f} B \xrightarrow{g} C$ is exact at B if $\operatorname{im} f = \ker g$. The weaker condition $g \circ f = 0$ only gives containment. The quotient $\ker g / \operatorname{im} f$ measures the difference. For a short exact sequence of finite-dimensional vector spaces, $\dim B = \dim A + \dim C$; the dimension identity alone does not specify the maps or prove exactness.

An exact sequence of modules splits when its quotient map has a module-linear section. A module P is projective if every map $P \to C$ lifts across every surjection $B \to C$. Equivalently every short exact sequence ending in P splits: apply the lifting property to 1_P, or form the pullback of a proposed map along the surjection. A module is projective if and only if it is a direct summand of a free module. For the nontrivial direction, choose a free surjection $F \to P$ and split it; for the converse, lift basis vectors of a free module and restrict the lift to the summand.

All vector spaces over k are free, hence projective. They are also injective: a map defined on a subspace extends by a basis extension. Consequently

$$\operatorname{Ext}_k^j(V,W) = 0, \qquad \operatorname{Tor}_j^k(V,W) = 0 \qquad (j > 0).$$

One may use the projective resolution concentrated in degree 0 to see both vanishings. They are statements about the coefficient ring k and its module category. They do not imply that every complex of vector spaces is exact, nor that $\operatorname{Ext}_{k[t]}$ or $\operatorname{Ext}_{k[G]}$ vanishes.

8.2 The smallest equivariant nonsplitting example

Let $R = k[t]$, $M = R/(t^2)$, and $K = R/(t)$. Define

$$0 \longrightarrow K \xrightarrow{i} M \xrightarrow{p} K \longrightarrow 0, \qquad i(a \bmod t) = at \bmod t^2, \quad p(f \bmod t^2) = f \bmod t.$$

The maps are well defined and R-linear; the image of i is exactly the kernel of p. As vector spaces, the sequence splits via $s(a) = a \bmod t^2$. As modules, it does not split. Any lift of $1 \in K$ has the form $1 + ct$ in M, and

$$t(1 + ct) = t \neq 0,$$

whereas $t \cdot 1 = 0$ in K. Thus no module-linear section exists. In the vector basis $(t, 1)$, multiplication by t has matrix $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Its invariant line kt has no invariant complement. This is the exact same obstruction expressed as a module extension, a failure of equivariant splitting, or a Jordan block.

The underlying vector-space splitting is useful, but forgetting the action has discarded the information responsible for the obstruction.

8.3 Extensions and the first Ext group, with an explicit computation

An extension of M by N is a short exact sequence $0 \to N \to E \to M \to 0$. Two extensions are equivalent when an isomorphism of their middle terms commutes with the specified inclusion and quotient maps and is the identity on N and M. Merely having isomorphic middle modules is a weaker relation. The equivalence classes form $\operatorname{Ext}_R^1(M,N)$ under Baer sum, with the split extension as zero. The general equivalence with the derived-functor definition is cited from Hilton–Stammbach III.1–2; we can construct the important cyclic case directly.

Let R be a commutative domain, $0 \neq f \in R$, and N an R-module. The free resolution

$$0 \longrightarrow R \xrightarrow{\times f} R \longrightarrow R/(f) \longrightarrow 0$$

is exact. Apply $\operatorname{Hom}_R(-,N)$. Evaluation at 1 identifies the two Hom modules with N, and the induced map is multiplication by f. Hence

$$\operatorname{Hom}_R(R/(f),N) = \ker(f : N \to N), \qquad \operatorname{Ext}_R^1(R/(f),N) \cong N/fN,$$

and the positive Ext groups of degree greater than 1 vanish for this first argument. The injectivity of $\times f$ uses both the domain and $f \neq 0$ hypotheses; over a ring with zero divisors this two-term complex may not be a resolution.

The extension meaning of N/fN can be seen without the derived-functor terminology. For an extension, choose $e \in E$ over 1 mod f. Then $fe = i(n)$ for a unique $n \in N$. Replacing e by $e + i(n_0)$ changes n by fn_0. Conversely, for $n \in N$ define

$$E_n = (N \oplus Re)/R(-n, fe),$$

meaning the relation $fe = n$. The copy of N injects because $rf = 0$ forces $r = 0$. The quotient is $R/(f)$; every extension has this form by expressing its elements using N and a lift of 1. Changes of the lift give precisely the equivalences $n \sim n + fn_0$. The zero class permits a lift killed by f, and therefore a section. Pullback followed by addition of the two copies of N sends classes $[n], [n']$ to $[n + n']$, explaining the addition law.

For $R = k[t]$ and $N = K = R/(t)$ this gives

$$\operatorname{Ext}^1_{k[t]}(K, K) \cong k.$$

The class $a \in k$ has generators u, e and relations $tu = 0$, $te = au$. If $a = 0$ it is split. For every $a \neq 0$, the middle module is isomorphic to $R/(t^2)$, but the extension classes for different a remain different with the endpoints fixed. Thus an extension invariant contains more information than an isomorphism class of a single operator.

8.4 Tor detects lost injections

Tensor the same free resolution with N. Its positive homology is

$$\operatorname{Tor}^R_1(R/(f), N) = \ker(f : N \to N), \qquad R/(f) \otimes_R N \cong N/fN.$$

The groups of degree greater than 1 vanish for this first argument. For $R = \mathbb{Z}$, $f = m \neq 0$, and $N = \mathbb{Z}/n$,

$$\mathbb{Z}/m \otimes \mathbb{Z}/n \cong \mathbb{Z}/\gcd(m, n), \qquad \operatorname{Tor}^{\mathbb{Z}}_1(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m, n).$$

For the kernel, multiplication by m on $\mathbb{Z}/n$ kills the subgroup generated by $n/\gcd(m, n)$, which has that order. This supplies a simple nonzero Tor example and explains the injection failure after tensoring the multiplication-by-m sequence.

An obstruction in Ext asks whether maps can lift or extend; an obstruction in Tor asks what tensoring has collapsed. They are useful descriptions only when the coefficient ring and the relevant sequence are retained.

8.5 Complexes and homology over a field

A chain complex is a sequence of maps $d_n : C_n \to C_{n-1}$ satisfying $d_{n-1} \circ d_n = 0$. Define

$$Z_n = \ker d_n, \qquad B_n = \operatorname{im} d_{n+1}, \qquad H_n(C) = Z_n/B_n.$$

A chain map $f : C \to D$ satisfies $d_n^D \circ f_n = f_{n-1} \circ d_n^C$. It carries cycles to cycles and boundaries to boundaries, hence induces $H_n(f)$. The quotient property constructs that map, and composition makes homology a functor.

Why the field does not force zero homology. The complex with a single nonzero term $C_0 = k$ has $H_0 = k$. In a two-term complex $0 \to k^2 \xrightarrow{d} k^2 \to 0$ with $d(x, y) = (x, 0)$, both H_1 and H_0 have dimension 1. All vector-space short exact sequences involved split; the map d still need not have maximal rank or have image equal to a prescribed cycle space.

In fact splitting gives a normal description of a complex. Choose complements

$$Z_n = B_n \oplus \mathcal{H}_n, \qquad C_n = Z_n \oplus L_n.$$

Then $\mathcal{H}_n \cong H_n(C)$ and $d_n : L_n \to B_{n-1}$ is an isomorphism: it is injective because $L_n \cap \ker d_n = 0$, and its image is all B_{n-1}. Thus

$$C_n = B_n \oplus \mathcal{H}_n \oplus L_n,$$

with differential zero on $B_n \oplus \mathcal{H}_n$ and an isomorphism from L_n onto B_{n-1}. A complex is therefore a direct sum of its homology with zero differential and two-term contractible pieces, after choices. The decomposition does not say the homology summands are zero.

More explicitly let $j : \mathcal{H} \to C$ and $q : C \to \mathcal{H}$ be inclusion and projection. Define $s_n : C_n \to C_{n+1}$ as the inverse of $d_{n+1} : L_{n+1} \to B_n$ on B_n, and zero on $\mathcal{H}_n \oplus L_n$. Then

$$d_{n+1}s_n + s_{n-1}d_n = 1_{C_n} - j_n q_n.$$

This is a chain homotopy from the identity to the projection onto chosen homology representatives. In particular an acyclic complex of vector spaces is contractible. Neither the contraction nor representatives are generally natural. Over $k[t]$, the required invariant complements can fail, so the same decomposition need not exist in that module category.

8.6 A connecting homomorphism is a controlled lifting construction

Let $0 \to A \xrightarrow{i} B \xrightarrow{p} C \to 0$ be a short exact sequence of chain complexes, exact in every degree. If $c \in C_n$ is a cycle, lift it to $b \in B_n$. Then $p(db) = dc = 0$, so uniquely $db = i(a)$ for $a \in A_{n-1}$. Since $i(da) = d^2 b = 0$, injectivity gives $da = 0$. Define

$$\delta : H_n(C) \to H_{n-1}(A), \qquad \delta([c]) = [a].$$

A different lift changes b by $i(a_0)$ and therefore changes a by da_0, a boundary. Changing c by a boundary and lifting its primitive changes b by a boundary in B, which does not alter a except for the preceding lift ambiguity. Thus δ is well defined and linear.

The resulting sequence

$$\cdots \to H_n(A) \to H_n(B) \to H_n(C) \xrightarrow{\delta} H_{n-1}(A) \to H_{n-1}(B) \to \cdots$$

is exact. For example, $\delta([c]) = 0$ means $a = da_0$; replacing b by $b - i(a_0)$ gives a cycle lifting c, so $[c]$ comes from $H_n(B)$. Conversely a cycle lift plainly gives $\delta([c]) = 0$. At $H_{n-1}(A)$, a cycle a dies in B precisely when $i(a) = db$; then $p(b)$ is a cycle and maps under δ to $[a]$. At $H_n(B)$, if $p(b) = dc'$ lift c' to b' and observe that $b - db'$ lies in $i(A)$ and is a cycle. These checks prove exactness at all three kinds of position.

This explains why exact sequences are not decorative notation: the connecting map measures the obstruction to lifting a cycle as a cycle.

8.7 A sensible stopping point for the homological toolkit

For ordinary finite-dimensional linear algebra, retain exactness, splitting, the explicit $k[t]/(t^2)$ example, complexes, and the one-step resolution calculations. Derived functors become worthwhile when a concrete functor fails to be exact or an equivariant extension remains nonsplit. General spectral sequences, derived categories, and long resolutions can be deferred until a specific problem needs them.

For the Hadamard/Williamson direction, a common-lift obstruction is not automatically an Ext class: sign constraints define a nonlinear finite set, not a module extension with an identified coefficient category. Calling it "homological" without constructing that category and obstruction would add a label without mathematical content.

Sources. Grillet X.1, X.3, XI.2, XI.6, and XII.1–5 give the general framework. Hilton–Stammbach III.1–4 (printed pp. 84–99; PDF pp. 95–110) is especially useful for extension equivalence and computations; IV.1–4 (printed pp. 117–130; PDF pp. 128–141) supplies complexes, connecting maps, homotopy, and resolutions. The cyclic extension construction and field-complex decomposition above are fully derived so the reader can use the ideas before reading a whole homological algebra text.

9 Representations and semisimplicity

9.1 An action becomes a module

A representation of a group G on a k-vector space is a homomorphism $\rho : G \to \mathrm{GL}(V)$. Extend linearly to the group algebra $k[G]$, whose elements are finite sums $\sum_g a_g g$ and whose product extends the group law bilinearly. Then

$$\left(\sum_g a_g g\right) \cdot v = \sum_g a_g \rho(g)(v)$$

makes V a $k[G]$-module. Conversely such a module restricts to a group action because g^{-1} supplies the inverse operator. Intertwining maps are exactly module maps, invariant subspaces are submodules, and irreducible representations are nonzero simple modules.

For the infinite cyclic group, $k[G] \cong k[t, t^{-1}]$; this describes an invertible operator. For a cyclic group of order m, $k[G] \cong k[t]/(t^m - 1)$; its representations are operators satisfying $T^m = 1_V$. If $\operatorname{char} k$ does not divide m, the polynomial $t^m - 1$ is square-free, since its derivative is mt^{m-1}. Over a field containing its roots, such operators are diagonalizable. Both separability and the splitting field enter this elementary representation example.

9.2 Maschke's averaging argument

Theorem. If G is finite and $|G|$ is invertible in k, every invariant subspace of a finite-dimensional representation has an invariant complement. Hence every finite-dimensional representation is a direct sum of irreducibles.

Proof. Let $U \subseteq V$ be invariant. Choose an ordinary linear projection $P : V \to U$ with $P|_U = 1_U$. Define

$$\overline{P} = \frac{1}{|G|} \sum_{g \in G} \rho(g) P \rho(g)^{-1}.$$

Each summand has image in U and is the identity on U, so the average does too; in particular $\overline{P}^2 = \overline{P}$. Reindexing the sum proves that $\overline{P}$ commutes with every $\rho(h)$. Therefore $\ker \overline{P}$ is an invariant complement. If V is not irreducible, split it and repeat, decreasing dimensions. □

The averaged projection depends on the original projection; the theorem asserts existence, not a preferred decomposition. Over $\mathbb{R}$ or $\mathbb{C}$, one can instead average a positive inner product. It remains positive because a nonzero vector has a positive contribution in each summand. Its invariant orthogonal complements prove complete reducibility with a geometric meaning.

Characteristic failure. In characteristic p, let a generator of C_p act on k^2 by $1 + N$, where $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Since $N^2 = 0$, $(1 + N)^p = 1$ in characteristic p (also when $p = 2$). This is a representation with invariant line ke_1 and no invariant complement, exactly as in the $k[t]/(t^2)$ example. Averaging would require $1/p$.

Finite-dimensional representations of infinite groups need not be semisimple even over $\mathbb{C}$: send $n \in \mathbb{Z}$ to $1 + nN$. The same invariant line fails to split. Finite-dimensionality alone does not remove extensions.

9.3 Schur's lemma and the role of the field

If S, T are irreducible representations, any intertwiner $f : S \to T$ is either zero or an isomorphism. Its kernel and image are invariant, and irreducibility forces the result. Thus $\operatorname{End}_G(S)$ is a division algebra over k.

If k is algebraically closed and S is finite-dimensional, every endomorphism f has an eigenvalue λ. The intertwiner $f - \lambda 1_S$ has nonzero kernel, so it is zero. Therefore $\operatorname{End}_G(S) = k1_S$. Neither the algebraic-closure nor finite-dimensional hypothesis should be suppressed.

Over $\mathbb{R}$, let C_4 act on the plane by a quarter-turn $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. There is no real invariant line, so the representation is irreducible, but its commuting endomorphisms include all $aI + bJ$, a copy of $\mathbb{C}$. Thus "irreducible implies scalar commutant" is false over $\mathbb{R}$.

For a finite-dimensional semisimple representation over an algebraically closed field, let S_i represent the distinct simple types occurring in V. The evaluation map gives the canonical isotypic description

$$\bigoplus_i S_i \otimes_k \operatorname{Hom}_G(S_i, V) \xrightarrow{\sim} V, \qquad s \otimes f \longmapsto f(s).$$

The group acts only on the first factor. To prove the map is an isomorphism, choose a decomposition $V \cong \bigoplus_i S_i^{m_i}$ and use Schur's lemma: the Hom space for S_i is k^{m_i} and contributes exactly that summand. The map was specified without this decomposition, so invertibility does not make the chosen copies canonical. The isotypic subspaces are intrinsic; a basis of each multiplicity space is not. Their commutant is a product of matrix algebras $\prod_i M_{m_i}(k)$ after choosing those bases.

9.4 Duals and tensors of representations

Define the dual action by

$$\rho^\vee(g)(\lambda) = \lambda \circ \rho(g^{-1}).$$

The inverse compensates for reversal under duality, ensuring $\rho^\vee(gh) = \rho^\vee(g)\rho^\vee(h)$. Plain transpose without the inverse would reverse the group multiplication. In dual bases its matrix is $\rho(g^{-1})^{\mathsf{T}}$, without complex conjugation.

For two representations, the diagonal tensor action is

$$g \cdot (v \otimes w) = \rho(g)(v) \otimes \sigma(g)(w).$$

Functoriality proves the group law. The canonical identification $W \otimes V^* \cong \operatorname{Hom}(V, W)$ carries the action to $f \mapsto \sigma(g) f \rho(g)^{-1}$. Its invariant vectors are the intertwiners. This is a practical use of tensor variance, not merely an alternative notation.

Characters are traces $\chi_V(g) = \operatorname{tr} \rho(g)$. Direct sums and tensor products satisfy

$$\chi_{V \oplus W} = \chi_V + \chi_W, \qquad \chi_{V \otimes W} = \chi_V \chi_W.$$

For finite groups over $\mathbb{C}$, characters determine isomorphism classes; the full orthogonality theorem proving that statement is a cited next step, not required to prove the two formulas. In modular characteristic the ordinary complex character package cannot be copied without changes.

9.5 From groups to Lie algebras: a short concrete bridge

For a differentiable representation of a matrix Lie group, differentiation at the identity yields a Lie algebra representation $d\rho : \mathfrak{g} \to \operatorname{End}(V)$ preserving brackets. Tensor actions differentiate according to the product rule:

$$X \cdot (v \otimes w) = (X \cdot v) \otimes w + v \otimes (X \cdot w).$$

Thus the Lie algebra operator is $A \otimes I + I \otimes B$, whereas the group operator is $\rho(g) \otimes \sigma(g)$. Confusing these two formulas changes the representation. Dualization differentiates to $X \cdot \lambda = -\lambda \circ d\rho(X)$.

Here is an explicit algebraic example over a characteristic-zero field. On homogeneous polynomials of degree m in x, y, set

$$E = x\frac{\partial}{\partial y}, \qquad F = y\frac{\partial}{\partial x}, \qquad H = x\frac{\partial}{\partial x} - y\frac{\partial}{\partial y}.$$

On the basis $v_j = x^{m-j}y^j$, $0 \le j \le m$,

$$Ev_j = jv_{j-1}, \qquad Fv_j = (m-j)v_{j+1}, \qquad Hv_j = (m-2j)v_j,$$

with missing endpoint vectors interpreted as zero. Direct calculation gives $[H, E] = 2E$, $[H, F] = -2F$, and $[E, F] = H$. This defines a representation of $\mathfrak{sl}_2$.

It is irreducible. The distinct eigenvalues $m - 2j$ of H allow polynomial projections onto its individual weight spaces by interpolation. A nonzero invariant subspace therefore contains some v_j. Repeated application of E reaches v_0, since the intervening integers are nonzero in characteristic zero; repeated application of F then reaches every v_j. The argument explicitly explains where small positive characteristic can fail.

Over $\mathbb{C}$, these are all finite-dimensional irreducible $\mathfrak{sl}_2$ representations. Hall §4.6 proves this classification using a highest-weight vector and the commutator identity $EF^jv = j(m-j+1)F^{j-1}v$ (printed pp. 96–101, PDF pp. 109–114). The notes above prove the model and its irreducibility, while the exhaustion statement is cited.

The associated $SU(2)$ representation is $\operatorname{Sym}^m \mathbb{C}^2$. The element $-I$ acts by $(-1)^m$, so it descends through $SU(2) \to SO(3)$ exactly when m is even. Integrating a Lie algebra representation and descending through a covering are extra global questions; Lie algebra isomorphisms alone do not identify the corresponding groups or their representations. Hall's group-versus-Lie-algebra sections and Xi III §3.6 are useful precisely at this boundary.

Selective sources. Xi III §3.2 gives an actual averaging proof (Theorem 3.20, printed pp. 109–111, PDF pp. 115–117), §3.7 constructs dual and tensor representations (pp. 152–162, PDF pp. 158–168), and §4.6 introduces $\mathfrak{sl}_2$ modules (starting p. 215, PDF p. 221). Grillet IX.2 and IX.7 place these in semisimple module theory. Hall §§4.3–4.7 gives the finite-dimensional Lie-theoretic version. Compact-group averaging requires integration and continuity, whereas the finite-group proof above requires only an invertible group order.

Terminology warning. Grillet IX.7 initially uses "irreducible" to mean indecomposable, that is, admitting no nontrivial invariant direct-sum decomposition (printed p. 381, PDF p. 393). He explicitly relates that definition to indecomposable modules on p. 383 (PDF p. 395). This study and Xi/Hall use the standard invariant-subspace definition, corresponding to a simple module. They coincide in a semisimple setting, but not for a nontrivial Jordan block. Thus Grillet IX.7.1, asserting a decomposition of every finite-dimensional representation into his "irreducibles," is not the complete-reducibility theorem with the terminology of these notes. This is a verified convention difference, not a mathematical counterexample to his stated definition.

10 Bridges to arithmetic, algebraic geometry, and differential topology

These sections explain how the same constructions lead outward. They are selective derivations and prerequisite diagnoses, not compressed substitutes for the destination books. The Hadamard/Williamson bridge follows separately and uses only the directions that have a concrete role there.

10.1 Finite extensions as linear algebra: trace and norm

Let L/k be a finite field extension of degree n. Every $a \in L$ defines the k-linear multiplication map

$$m_a : L \to L, \qquad m_a(x) = ax.$$

Define $\mathrm{Tr}_{L/k}(a) = \mathrm{tr}(m_a)$ and $N_{L/k}(a) = \det(m_a)$. Since $m_{ab} = m_a \circ m_b$ and $m_{a+b} = m_a + m_b$, norm is multiplicative and trace is k-linear. For $c \in k$, $m_c = c1_L$, so trace is nc and norm is c^n. The map $a \mapsto m_a$ is faithful: evaluate at 1 to recover a.

For $L = k(\alpha)$ with monic minimal polynomial $f(t) = t^n + a_{n-1}t^{n-1} + \cdots + a_0$, multiplication by α has the companion matrix of f. Hence

$$\mathrm{Tr}_{L/k}(\alpha) = -a_{n-1}, \qquad N_{L/k}(\alpha) = (-1)^n a_0.$$

If L/k is separable, let Ω be an algebraic closure of k. The primitive-element theorem writes $L = k(\alpha)$; it is cited here, not assumed for inseparable extensions. The polynomial f has distinct roots $\alpha_1, \dots, \alpha_n$, and

$$\Omega \otimes_k L \cong \Omega[t]/(f) \cong \prod_{i=1}^{n} \Omega$$

by scalar extension and Chinese remaindering. Multiplication by a becomes diagonal with entries $\sigma_i(a)$ for the n embeddings $\sigma_i : L \to \Omega$. Trace and determinant are unchanged by scalar extension of the matrix, so

$$\mathrm{Tr}_{L/k}(a) = \sum_i \sigma_i(a), \qquad N_{L/k}(a) = \prod_i \sigma_i(a).$$

This is the same polynomial-splitting mechanism used for operators. The new content is that the regular representation comes from a field multiplication.

The trace pairing $b(a, b) = \mathrm{Tr}_{L/k}(ab)$ is nondegenerate for finite separable extensions. In the power basis its Gram matrix is $V^{\mathsf{T}}V$, where $V_{ij} = \alpha_i^{j-1}$. The Vandermonde determinant is nonzero because the roots are distinct; therefore its square, the determinant of the trace Gram matrix, is nonzero. This proves nondegeneracy even in positive characteristic when the extension is separable; positivity has no role.

For a finite purely inseparable counterexample, take $k = \mathbb{F}_p(s)$ and $L = k(\alpha)$ with $\alpha^p = s$. In the basis $1, \alpha, \dots, \alpha^{p-1}$, multiplication by $\sum_{j=0}^{p-1} a_j \alpha^j$ has every diagonal entry a_0; the other powers produce

cyclic shifts. Thus its trace is $pa_0 = 0$. The trace pairing is identically zero although $[L : k] = p$. A field extension being finite does not suffice for trace duality.

Xi III §5.8 (printed pp. 272–278, PDF pp. 278–284) is the immediate Galois-theoretic source. Lang III.1 (printed pp. 57–62, PDF pp. 71–76) uses the separable trace pairing to define complementary integral modules. The latter retains an integral lattice, which is extra information beyond the k-vector space and its nondegenerate pairing.

10.2 Why arithmetic needs more than field extensions

In a number field, the ring of integers is a $\mathbb{Z}$-module inside a $\mathbb{Q}$-vector space. Tensoring with $\mathbb{Q}$ recovers the ambient vector space but loses torsion and integral index. For example $2\mathbb{Z} \subset \mathbb{Z}$ becomes an isomorphism after tensoring with $\mathbb{Q}$, while its cokernel is $\mathbb{Z}/2$ before tensoring. Integral bases and discriminants cannot be read from the rational vector space alone.

Localizing a commutative ring R at a multiplicative set S produces $S^{-1}R$ with a map from R making all elements of S invertible. It is universal for that property. For modules, $S^{-1}M \cong S^{-1}R \otimes_R M$ and localization is exact. For instance if a localized element maps to zero, some denominator in S annihilates its original image; multiplying the numerator by that denominator reduces the kernel assertion to the original exact sequence. This proves the middle exactness directly, with surjectivity obtained by lifting numerators.

A finitely generated torsion-free module over a Dedekind domain is projective (Lang I.9, Proposition 26, printed p. 29, PDF p. 43). It need not be free: a nonprincipal invertible ideal is a rank-one example. Its scalar extension to the fraction field is one-dimensional, so vector-space dimension misses its ideal class. Over a PID, this distinction vanishes for finitely generated torsion-free modules. That is the arithmetic boundary of the elementary structure theorem.

To read Lang efficiently, the missing bridges are integral dependence and integral closure, prime ideal factorization in Dedekind domains, discrete valuations, completions, and local ramification. Some algebraic ingredients appear in Hungerford VIII and Grillet's commutative algebra chapters; their existence in a table of contents does not establish complete preparation for Lang's local and analytic arguments. Lang I.9 explicitly says it is not used later in that book, so projectivity should be read for this conceptual bridge, not falsely advertised as a prerequisite for all his subsequent chapters.

10.3 Affine geometry: symmetric algebra, ideals, and tangent vectors

For a finite-dimensional vector space V, the affine coordinate algebra is $\mathrm{Sym}(V^*)$. Imposing polynomial equations amounts to taking a quotient by an ideal. Over an algebraically closed field, passage between radical ideals and algebraic sets requires the Nullstellensatz; it is a substantial theorem, not an immediate consequence of the quotient property. Nonradical ideals retain nilpotent information invisible to a set of points.

For a finitely generated k-algebra A and a k-rational point $x : A \to k$, put $\mathfrak{m} = \ker x$. A tangent vector at x can be defined as a k-linear derivation $D : A \to k$ satisfying

$$D(ab) = x(a)D(b) + x(b)D(a).$$

It vanishes on constants and on $\mathfrak{m}^2$. Conversely every functional $\ell : \mathfrak{m}/\mathfrak{m}^2 \to k$ defines $D(a) = \ell((a - x(a)) \bmod \mathfrak{m}^2)$. Expanding a product proves the derivation law, and these assignments are inverse. Thus

$$T_x \operatorname{Spec} A \cong \operatorname{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, k).$$

This statement uses a rational point, so its residue field is k. At a general scheme point the residue field and the relative differential module must be tracked.

For $A = k[x_1, \dots, x_n]/I$ at a rational point a, expand every $f \in I$ modulo the square of the maximal ideal:

$$f(a+u) \equiv f(a) + \sum_i \frac{\partial f}{\partial x_i}(a) u_i \pmod{(u)^2}.$$

The tangent space is the common kernel of these linear equations. For the cusp $y^2 = x^3$ at the origin, there is no nonzero linear equation from its defining polynomial, so the tangent space is two-dimensional even though the curve has dimension one. For $A = k[\varepsilon]/(\varepsilon^2)$, the point has a one-dimensional tangent space despite its underlying point set having one point. Nilpotents store infinitesimal directions.

Tensor products now do geometric work: affine fiber products are represented by $A \otimes_R B$. Fibers of a module use $M \otimes_R \kappa(\mathfrak{p})$. Because this tensor operation may fail to preserve injections, flatness is needed to control families. The finite-dimensional vector-space version of exactness is insufficient once the coefficient ring varies.

10.4 What the scheme and sheaf bridge adds

Hartshorne II.1–2 introduces sheaves and schemes; II.5 develops sheaves of modules, and II.8 introduces differentials. These topics add local data and gluing, not just new names for global vector spaces. A locally free sheaf is locally a finite direct sum of the structure sheaf; a global free basis is an extra condition. On an affine scheme, finite locally free sheaves correspond to finitely generated projective modules, a theorem requiring the module–sheaf correspondence and its local criteria.

Global sections form a left-exact functor; their higher derived functors are sheaf cohomology. The obstruction here concerns sheaves and gluing even when the ground field is a field. It is not contradicted by $\operatorname{Ext}^j_k(V, W) = 0$: these are different categories and different functors. Hartshorne III.1–3 develops that transition. His introduction marks externally cited commutative-algebra and homological results with an "A" instead of proving them all in place (printed pp. xv–xvi, PDF pp. 14–15). The collection provides many algebraic ingredients, but a reader still needs localization, local rings, sheaves and stalks, and substantial commutative-algebra fluency before this becomes a comfortable route.

10.5 Differential topology: linear algebra varies smoothly

For a smooth map $F : M \to N$, differentiation gives maps $dF_x : T_xM \to T_{F(x)}N$. The chain rule makes differentiation functorial. At a regular point of a level set $F^{-1}(y)$, the implicit-function theorem identifies

$$T_x F^{-1}(y) = \ker dF_x.$$

For a smooth embedded submanifold $M \subset N$, there is an exact sequence of vector bundles over M,

$$0 \to TM \to TN|_M \to \nu \to 0,$$

where ν is the quotient normal bundle. A Riemannian metric on N identifies it with the orthogonal normal bundle. On a paracompact smooth base, a smooth positive bundle metric exists by partitions of unity, so every short exact sequence of smooth vector bundles splits. The splitting need not be natural without the chosen metric or other extra data.

This is stronger than choosing complements separately at each point: arbitrary pointwise choices need not vary continuously. Conversely splitting does not make either summand trivial. A nontrivial line bundle can be a direct summand of a trivial higher-rank bundle. Smooth variation and a global frame are different requirements.

Hirsch proves the bundle splitting theorem as IV.2.2 (printed p. 95, PDF p. 105), using partitions of unity, and defines the orthogonal structure immediately afterward. This is actual proof coverage, not an analogy inferred from the chapter title. The necessary new bridges for his development are multivariable calculus, the inverse/implicit-function theorems, general topology, manifolds, partitions of unity, and then approximation and transversality.

The Hessian of a smooth real function at a critical point is a symmetric bilinear form on the tangent space. Its inertia survives coordinate change, which explains why Sylvester's law underlies the Morse index. The Morse lemma and Morse theory additionally require analytic and topological arguments; diagonalization alone does not prove their local normal form or global conclusions. Hirsch VI is a destination for those arguments.

Exterior algebra can build differential forms once smooth bundles exist, but Hirsch's preface expressly chooses a route without tensors or differential forms. It would be misleading to force homological algebra, algebraic geometry, or tensor calculus into his prerequisite chain. The common foundation is careful linear geometry; the subsequent methods are genuinely additional.

11 A concrete destination: Hadamard and Williamson foundations

11.1 Scope and the exact question

The confirmed existing direction is the paused Hadamard/Williamson project in this repository. Its general existence question asks whether, for every positive integer k, there is a matrix

$$H \in \{-1, 1\}^{4k \times 4k}, \qquad HH^{\mathsf{T}} = 4kI_{4k}.$$

The prior work at commit `2834fea31fb4ca7203531ca220e55a2fb7640119` records three rounds of bounded research in a restricted family. Its proofs and limitations are read here as the starting point. No prior benchmark is presented as a fresh run, and no historical claim about the smallest missing order or the worldwide current status is imported without a separate literature audit. This chapter does not restart the computational search.

The structural chain actually needed is

$$\text{Gram matrices} \longrightarrow \text{tensor constructions}, \qquad \text{finite fields} \longrightarrow \text{quadratic characters},$$

cyclic group algebras ⟶ Fourier decomposition ⟶ correlation equations and lifting constraints.

It makes direct use of Xi II's metric and tensor material and Xi III's groups, representations, and finite fields. General Ext theory, schemes, and differential topology have no established role in this chain.

11.2 Gram matrices and the order restriction

The identity $HH^{\mathsf{T}} = nI$ says precisely that the rows are mutually orthogonal in the usual positive real inner product and have squared length n. It also gives $H^{-1} = n^{-1}H^{\mathsf{T}}$ and $|\det H| = n^{n/2}$, attaining the determinant bound proved in the forms chapter. Orthogonality alone does not enforce sign entries; both conditions are essential.

Multiplying rows or columns by signs and permuting them preserves the Hadamard condition. Normalize the first row to all $+1$. A second row must have equally many signs, so n is even. When $n > 2$, permute its positive entries into the first half and let a third row have a positives in the first half and b in the second. Its products with the first two rows give

$$2(a + b) = n, \qquad 2(a - b) = 0.$$

Hence $n = 4a$. This proves the necessary orders $1, 2$, or a multiple of 4. It supplies no construction at each admissible order. The proof is over the integers or reals; reducing the equation modulo 2 would collapse all signs and lose the argument.

11.3 Tensor closure and what it leaves unsolved

If H and K have orders a, b, functoriality of the tensor product and transpose gives

$$(H \otimes K)(H \otimes K)^{\mathsf{T}} = (HH^{\mathsf{T}}) \otimes (KK^{\mathsf{T}}) = abI_{ab}.$$

Its entries remain signs. Tensoring with $S = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ gives $\begin{pmatrix} H & H \\ H & -H \end{pmatrix}$ and yields all powers-of-two orders from [1]. This is a genuine infinite construction theorem. It does not produce all multiples of four without additional seeds or constructions. The universal tensor identities explain why the construction transports orthogonality, while the sign-entry requirement explains why arbitrary orthogonal matrices are not sufficient seeds.

11.4 Finite fields and a character identity proved by counting

Let $\mathbb{F}_q$ be a finite field of odd order. Its nonzero squares form an index-two subgroup: the squaring map has exactly two preimages $x, -x$ for each nonzero square. Define the quadratic character $\chi : \mathbb{F}_q \to \{0, \pm 1\}$ with $\chi(0) = 0$, $+1$ on nonzero squares, and -1 on nonsquares. The quotient-group description proves multiplicativity, and equal coset sizes prove $\sum_x \chi(x) = 0$.

The multiplicative group of a finite field is cyclic. One short proof lets e be the least common multiple of its element orders. In a finite abelian group, combine elements of maximal prime-power orders to obtain an element of order e. Every nonzero field element is a root of $X^e - 1$, so there are at most e of them, whereas Lagrange gives $e \le q - 1$. Thus $e = q - 1$. It follows that $\chi(-1) = (-1)^{(q-1)/2}$.

For $a \neq 0$, the crucial correlation identity is

$$\sum_{x} \chi(x)\chi(x+a) = -1.$$

Count pairs (x, y) with $y^2 = x(x+a)$. For fixed x, their number is $1 + \chi(x(x+a))$, so the total is q plus the displayed sum. Scale $x = az$, $y = aw$ and rewrite as

$$(2z + 1 - 2w)(2z + 1 + 2w) = 1.$$

Since 2 is invertible, each nonzero first factor determines a unique second factor and a unique (z, w). There are $q - 1$ pairs, proving the sum. For $a = 0$ the corresponding sum is $q - 1$. This argument uses a finite field of odd order, not the special fact that its elements are residue classes modulo a prime.

11.5 Paley constructions with explicit signs

Index Q by $x, y \in \mathbb{F}_q$ and put $Q_{xy} = \chi(y - x)$. The character identities give

$$Q\mathbf{1} = 0, \qquad QQ^{\mathsf{T}} = qI - J, \qquad Q^{\mathsf{T}} = \chi(-1)Q.$$

If $q \equiv 3 \pmod 4$, set

$$H = \begin{pmatrix} 1 & \mathbf{1}^{\mathsf{T}} \\ \mathbf{1} & Q - I \end{pmatrix}.$$

Its entries are signs, its top row is orthogonal to every lower row because $1 + (Q - I)\mathbf{1} = 0$, and its lower Gram block is

$$J + (Q - I)(Q^{\mathsf{T}} - I) = J + qI - J - Q - Q^{\mathsf{T}} + I = (q + 1)I.$$

Therefore $HH^{\mathsf{T}} = (q + 1)I$. The prior study used the equivalent sign convention with lower first column $-\mathbf{1}$ and core $Q + I$. The conventions have been derived separately here so a copied sign does not silently spoil orthogonality.

For $q \equiv 1 \pmod 4$, define the symmetric conference matrix

$$C = \begin{pmatrix} 0 & \mathbf{1}^{\mathsf{T}} \\ \mathbf{1} & Q \end{pmatrix}.$$

It has zero diagonal, off-diagonal signs, and $C^2 = qI$. The block matrix

$$\mathcal{H} = \begin{pmatrix} C + I & C - I \\ C - I & -C - I \end{pmatrix}$$

has only sign entries. The diagonal Gram block is $(C+I)^2 + (C-I)^2 = 2(q+1)I$ and the off-diagonal block vanishes by commutation. It is Hadamard of order $2(q + 1)$.

These proofs add a short prime-power and conference-matrix development to the prior study's prime, $q \equiv 3 \pmod 4$ proof. They are classical constructions, not new results or claims that the earlier scripts implemented them. Finite fields of every prime-power order exist (Xi III Theorems 5.40–5.45, printed pp. 260–261, PDF pp. 266–267); these pages and their proofs were inspected visually. Small exact instances are checked separately in this deliverable, without resuming an enumeration campaign.

11.6 Circulants as an integral group algebra

Fix $m \geq 1$ and let P be the cyclic shift with $P_{ij} = 1$ when $j = i + 1 \pmod m$. The map

$$\mathbb{Z}[z]/(z^m - 1) \longrightarrow M_m(\mathbb{Z}), \qquad f \longmapsto f(P)$$

is injective: the first row of $\sum_{j=0}^{m-1} a_j P^j$ is its coefficient vector. Its image is the circulant algebra, so circulants commute. Define $f^\iota(z) = f(z^{-1})$. Transpose corresponds to this involution. For a real sequence x with polynomial $X(z) = \sum_j x_j z^j$, the coefficient of z^s in XX^ι is

$$\mathrm{PAF}_x(s) = \sum_j x_j x_{j+s}.$$

This follows by collecting terms with exponent difference s; autocorrelation is even in s. It is convolution with the reversed sequence, so the group algebra retains the integral coefficients needed for exact verification.

If $x_j = x_{-j}$, then $X = X^\iota$ and the circulant is symmetric. Four such sign sequences form a Williamson quadruple when

$$AA^\mathsf{T} + BB^\mathsf{T} + CC^\mathsf{T} + DD^\mathsf{T} = 4mI_m,$$

equivalently their four autocorrelations sum to zero at every nonzero shift. Symmetry permits replacing the four Gram matrices by A^2, B^2, C^2, D^2. Commutation and symmetry then make

$$W = \begin{pmatrix} A & B & C & D \\ -B & A & -D & C \\ -C & D & A & -B \\ -D & -C & B & A \end{pmatrix}$$

Hadamard of order $4m$. Multiplying blocks, every diagonal Gram block is the sum of the four squares, while an off-diagonal block has terms such as $-AB + BA - CD + DC = 0$. Transpose and commutation hypotheses are indispensable in that cancellation. Existence in the Williamson family is sufficient for a Hadamard matrix of that order; it is not a necessary characterization of all Hadamard matrices.

11.7 Fourier characters are Chinese remainder coordinates

Let k have characteristic not dividing m and contain a primitive mth root ζ. The roots of $z^m - 1$ are distinct, so

$$k[z]/(z^m-1) \xrightarrow{\sim} k^m, \qquad f \longmapsto (f(1), f(\zeta), \dots, f(\zeta^{m-1}))$$

is a Chinese remainder isomorphism. Its inverse is the finite Fourier formula

$$a_j = \frac{1}{m}\sum_{r=0}^{m-1} f(\zeta^r)\zeta^{-rj}.$$

To verify it, the geometric sum $\sum_r \zeta^{r\ell}$ equals m when $m \mid \ell$ and 0 otherwise. In the regular representation, the vector $(1, \zeta^r, \dots, \zeta^{(m-1)r})$ is an eigenvector of P with eigenvalue ζ^r. Thus all circulants diagonalize in the same basis; this is the group representation decomposition into characters.

Over $\mathbb{C}$, divide these vectors by $\sqrt{m}$ to get a unitary basis. A real symmetric circulant then has real eigenvalues, since $X(\zeta^{-r}) = X(\zeta^r) = \overline{X(\zeta^r)}$. The correlation identity becomes

$$\sum_{X\in\{A,B,C,D\}} |X(\zeta^r)|^2 = 4m \qquad (0 \le r < m).$$

The frequency-zero equation is the row-sum identity $r_a^2 + r_b^2 + r_c^2 + r_d^2 = 4m$. A single row or partial tuple must satisfy the corresponding upper bounds, but separate feasible spectral values need not come from any common sign coefficients.

The field assumptions explain failures: over $\mathbb{Q}$ the algebra need not split into one-dimensional factors, although it splits after adjoining ζ; if the characteristic divides m, $z^m - 1$ has repeated roots and this semisimple Fourier argument breaks. Positivity and absolute-square inequalities require the real/complex setting; they have no automatic meaning over a finite field.

11.8 A quotient test is not a lift theorem

For $d \mid m$, the group homomorphism $C_m \to C_d$ gives an algebra quotient sending z to the generator of order d. The compressed coefficient is $y_r = \sum_{j\equiv r \ (d)} x_j$. Applying the quotient to the four correlation identity gives

$$\sum_x C_{y_x} C_{y_x}^{\mathsf{T}} = 4mI_d.$$

The right side remains $4m$, not $4d$, and compressed coefficients are integer sums rather than signs. This is a necessary consequence of a full solution. The quotient discards coefficients, so its converse needs a separate common-lift theorem.

The prior proof gives an exact illustrative failure at $m = 9$. The normalized symmetric masks $(2, 10, 10, 13)$, where bit $j-1$ negates positions $\pm j$, compress at $d = 3$ to

$$(3,1,1), \quad (3,-1,-1), \quad (3,-1,-1), \quad (-1,-1,-1).$$

Their total compressed autocorrelation is $(36, 0, 0)$ and their row-square sum is 36, so both proper cyclic quotients of C_9 pass. But their uncompressed autocorrelation at shift 1 sums to -8. This shows that these particular lifts fail; it alone would not prove that every other lift of the compressed

multiset fails. The prior independent review separately records a stronger no-common-lift fixture: the compressed multiset

$$(-1,-1,-1),\quad (-1,1,1),\quad (-1,3,3),\quad (3,-1,-1).$$

It too has total compressed autocorrelation $(36,0,0)$. There are 16 normalized symmetric sign rows at core nine; among their four-row multisets, exactly 27 map to this specified compressed multiset. The new exact checker independently reconstructs and rejects all 27 by their full autocorrelations. Thus no common lift exists **in this normalized symmetric Williamson family**. That is stronger evidence than one failed lift, and still says nothing about nonexistence of arbitrary Hadamard matrices of order 36.

Even positive spectral residuals need not have a sign lift. At $m = 7$, let three sequences have negative positions $\{\pm1\}$, $\{\pm2\}$, and $\{\pm3\}$. Their squared circulants sum to $20I + J$, leaving $8I - J$ toward the target $28I$. This residual is positive definite: its eigenvalues are 1 on constants and 8 on their orthogonal complement. Nevertheless the odd Williamson product constraint below forces the fourth sequence to be all positive; its row-sum square is 49, while the remaining row-square budget is 1. No fourth Williamson row exists. A positive real Gram factor does not necessarily satisfy the required discrete sign pattern.

11.9 Why four colors appear in the prior reduction

Now let $m = 2h + 1$ be odd and normalize every sequence by $x_0 = 1$. Write $T(z) = \sum_{j=0}^{m-1} z^j$ and $N_x(z) = \sum_{x_j=-1} z^j$, so $X = T - 2N_x$. The identities $T^2 = mT$ and $TN_x = |N_x|T$, substituted into the Williamson equation and divided coefficientwise by 4 in the integral group algebra, give

$$\sum_x N_x^2 + (m - R)T = m, \qquad R = \sum_x |N_x|.$$

Each $|N_x|$ is even by symmetry and normalization. Modulo 2, squaring removes cross terms and sends z^j to z^{2j}. Since doubling permutes residues modulo odd m, each nonzero position contains an odd number of negative entries among the four rows. Equivalently

$$a_j b_j c_j d_j = -1 \qquad (j \neq 0).$$

This is the odd Williamson product theorem in the prior proof. Its hypotheses are structural restrictions; it is not a constraint on arbitrary Hadamard matrices.

Let $v_c = \mathbf{1} - 2e_c \in \mathbb{Z}^4$, $c = 1, 2, 3, 4$. The eight four-entry sign columns with product -1 are uniquely εv_c with $\varepsilon = \pm1$, and

$$v_c^{\mathsf{T}} v_d = 4[c = d], \qquad \mathbf{1}^{\mathsf{T}} v_c = 2.$$

Thus each nonzero column has a color κ_j and sign ε_j, reflected at opposite indices. For a nonzero shift t, pair the correlation terms indexed by j and $-t - j$. The pair involving zero contributes $4\varepsilon_t$; the unique fixed term contributes 4. The remaining $h - 1$ pairs define edges E_t on the opposite-index classes. Each contributes $8\varepsilon_u\varepsilon_v[\kappa_u = \kappa_v]$. Hence exact correlation is equivalent to

$$\varepsilon_t + 1 + 2 \sum_{\{u,v\} \in E_t} \varepsilon_u \varepsilon_v [\kappa_u = \kappa_v] = 0.$$

If c_t counts its monochromatic edges, reducing this equation modulo 4 gives $\varepsilon_t = -(-1)^{c_t}$. All signs are therefore forced by a complete coloring. The full integer equations still have to hold; the parity consequence alone is insufficient.

Permuting the four block rows merely permutes color names. Naming colors in order of first appearance, with at most four colors, gives a unique restricted-growth representative of each orbit. This is a quotient by block permutations, including repeated blocks. It is not classification up to all Hadamard matrix equivalences, nor a quotient by arbitrary coordinate changes.

11.10 Boolean polynomials and the precise relaxation boundary

Encode each color minus one by two bits a_i, b_i. In the Boolean algebra

$$B = \mathbb{F}_2[a_i, b_i \ (1 \le i \le h)]/(a_i^2 - a_i, b_i^2 - b_i),$$

every function on the bit cube has a unique squarefree polynomial. Induct on the last bit: any function is $f_0 + x(f_1 - f_0)$, which proves existence and uniqueness recursively. Equality of colors has polynomial

$$\delta_{uv} = (1 + a_u + a_v)(1 + b_u + b_v).$$

It is 1 exactly when both bits agree. Therefore

$$n_t = 1 + \sum_{\{u,v\} \in E_t} \delta_{uv}, \qquad \varepsilon_t = (-1)^{n_t}$$

has degree at most two in the bits. Substituting a partial assignment preserves this degree bound. Write a residual $n_i = \alpha_i + p_i$ with constant α_i and zero-constant polynomial p_i, and put $\eta_i = (-1)^{\alpha_i}$, $\xi_p = (-1)^p$. Then

$$\varepsilon_i = \eta_i \xi_{p_i}, \qquad \varepsilon_u \varepsilon_v = \eta_u \eta_v \xi_{p_u + p_v}.$$

These are exact identities of functions. Equal residual polynomials therefore represent the same unknown phase even before its sign is known. Summing their integer coefficients first can reveal cancellation that separate sign ranges miss. Constants and linear terms must be included when deciding equality; equal quadratic parts alone do not suffice.

The exponents are computed in B, of characteristic two. The correlation coefficients and their sums are computed in $\mathbb{Z}$. The map $p \mapsto (-1)^p$ converts addition of Boolean functions into multiplication of signs; it is not a ring map allowing integer coefficients to be reduced modulo two.

After grouping, a lag expression can have the exact form $K + \sum_p c_p \xi_p$. Treating distinct ξ_p as independent signs gives a safe superset of its attainable values, hence a sound rejection test when zero is absent. Presence of zero is only a necessary condition. Relations such as $\xi_{p+q} = \xi_p \xi_q$ remain; even preserving all such linear phase relations need not enforce the quadratic origin. For example,

the values of a, b, ab cannot be $(0, 0, 1)$, though these three polynomials are linearly independent over $\mathbb{F}_2$. A prospective stronger method must specify which extra dependencies it keeps and prove its rejection rule.

The prior work's core-11 false-positive prefix is a concrete warning that separately feasible row and lag relaxations can lack a common completion. Its existing audit, not a new run here, supports that bounded observation. Neither the algebraic reduction nor the word "exact" makes every downstream relaxation complete.

11.11 What to learn next for this direction

Tool needed for the study	Source in the collection	Supplement supplied or still needed
Orthogonality, Gram matrices, positive residuals	Xi II §§3.1–3.5, 3.8	Gram-factor versus sign-factor distinction; examples above
Kronecker products and tensor identities	Xi II §6.2; Hungerford IV.5; Grillet XI.5–6	Sylvester closure proof above
Cyclic groups, quotient rings, polynomial actions	Xi III §§1.2, 4.2–4.4; Hungerford III–IV; Grillet III, VIII	Integral cyclic-algebra involution and correlation dictionary above
Finite fields and multiplicative characters	Xi III §5.6; Hungerford V.5; Grillet IV.1.6 and V.1.3	Quadratic-character sum and both Paley block proofs above
Character decomposition and invariant splitting	Xi III §§3.1–3.5; Grillet IX.2, IX.7–9	Fourier inversion as CRT, with splitting/characteristic hypotheses above
Compression and common lifting	Quotients and maps throughout the core	A real additional theorem would have to couple sign lifts; none is assumed
Quadratic Boolean constraints	Polynomial rings, quotients, functions on finite fields	Boolean normal form and phase dependence above; algorithmic feasibility remains a separate subject

The strongest useful next reading task is to reconstruct each displayed dictionary and counterexample, then read the pinned prior proofs with the coefficient rings visible. A later research proposal should target a specific lost dependency, a genuinely wider construction family, or a proved lift criterion. Such proposals are prospective; no improvement in search complexity or progress toward universal existence is claimed by this foundational chapter.

12 Selective comparison: what can be reorganized, what must be retained

12.1 How to read the comparison

The criterion is the user's intended use: rebuilding algebra around reusable principles, then returning to linear algebra and the Hadamard/Williamson study. "Covered" has several meanings. A theorem may be stated, proved through a different construction, left as an exercise, or illustrated geometrically. These are distinct contributions. A chapter list can establish where to look; it cannot establish that two books are interchangeable.

Evidence levels and edition/page conversions are in the source ledger. The comparisons below are

selective, and pedagogical judgments are identified as judgments based on the inspected material. No global difficulty ranking or exercise-by-exercise replacement audit is claimed.

12.2 Xi I versus the graduate algebra books

Xi I starts with equations and low-order determinants, develops sets and maps, then connects vector rows/columns, rank, and linear maps before abstract group/ring/field language. Its Chapter 4 does more than provide a determinant formula: oriented-volume motivation and multilinearity appear on printed pp. 75–76 (PDF pp. 81–82), and §4.4 gives multiple characterizations on p. 93 (PDF p. 99). The surrounding exercises include minor identities. That material has structural value even when its elementary calculations are already familiar.

Hungerford III and Grillet III give a broader ring-theoretic organization of polynomial division and factorization; their later module chapters explain why a PID is the right setting for one-operator structure. The present study therefore reorganizes the **use** of Xi I's polynomial arithmetic into Bézout, Chinese remainder, and annihilator arguments. It does not infer that Xi I's derivations, examples, or root-location exercises are redundant.

Xi I §7.4 starts with symmetry in roots and elementary symmetric polynomials (p. 177, PDF p. 183); §7.6 makes common-factor detection into a resultant matrix (p. 190, PDF p. 196). These are useful links from coefficient calculations to invariant expressions and elimination. Grillet IV.7's resultant/discriminant location is verified from contents (p. 176, PDF p. 188), but its entire proof and exercise treatment was not compared against Xi's. No preference about replacement follows from that heading alone.

For a returning reader who can prove quotient factorization, rank–nullity, determinant multilinearity, Euclidean division, and Bézout, much of Xi I can be used as a reference rather than reread sequentially. That is a competence criterion for selective study, not a judgment that introductory material lacks content.

12.3 Xi II and two routes to Jordan form

There is substantial, verified theorem overlap. Xi II Theorem 2.51 (printed pp. 72–73, PDF pp. 78–79) proves the coprime-annihilator split using Bézout. Theorems 2.52–2.53 build the generalized eigenspaces. Theorems 2.56–2.57 then establish Jordan form under the actual hypothesis that the characteristic polynomial splits in the base field. Existence is proved by quotient induction for nilpotent operators; uniqueness comes from dimensions of images of powers (pp. 75–77, PDF pp. 81–83). These proof pages were read, not inferred from the section title.

Grillet VIII.6 proves the PID structure theorem (pp. 336–342, PDF pp. 348–354) and VIII.7 makes V a $k[t]$-module, proves its annihilator has a monic generator, and derives primary cyclic blocks and Jordan bases (pp. 342–346, PDF pp. 354–358). Its sampled proof gives a real alternative construction, with rational canonical forms and general coefficient rings naturally close at hand. Hungerford's suggested linear-algebra course explicitly takes torsion PID modules before VII.1–5 (printed p. xviii, PDF p. 15); his IV.6 and VII.4–5 locations support the same broad organization. A detailed proof-by-proof audit of Hungerford VII was not made.

For the user's purpose, the module route is an efficient **structural spine**: one remembers a module and its annihilator, then recovers the canonical forms. Xi's route retains pedagogical value: it shows what the projections do inside the actual vector space and constructs chains through quotients. A

reader who knows only the abstract theorem should still reconstruct one chain and the kernel-growth count. The two proofs illuminate different failure cases.

12.4 Xi II tensor constructions versus Grillet and Mac Lane

Xi II Definition 6.11 constructs tensor products from a basis criterion, proves uniqueness, and **proves the universal property** in Theorem 6.15 (pp. 273–275, PDF pp. 279–281). It is inaccurate to say that this book has only coordinates or no universal-property treatment. Its Hom identification, scalar extension, functorial maps, Kronecker products, contraction as trace, and contraction as operator composition are explicit examples (pp. 274–283, PDF pp. 280–289).

Grillet XI.5 constructs tensor products by generators and relations for a right module and a left module, proving the balanced universal property (pp. 434–437, PDF pp. 446–449). XI.6–8 adds right exactness, dual modules, and flatness. This is a genuine extension of scope: vector-space bases do not exist for arbitrary modules, and tensoring can destroy injections. The simplest $\mathbb{Z}/2$ counterexample explains why that generality matters.

Mac Lane III.1–2 (pp. 55–62, PDF pp. 68–75) identifies universal arrows, universal elements, and representability, and proves Yoneda. IV.1 (pp. 79–80, PDF pp. 92–93) formulates adjunction, including the Hom–tensor example. Its contribution is a common recognition language, with careful naturality. It does not replace tensor calculations or tell the reader which finite-dimensionality hypotheses can be forgotten.

There is one confirmed scope error: the remark after Xi II Theorem 6.19 (p. 278, PDF p. 284) extends its identification of **all** mixed multilinear forms to infinite-dimensional spaces. The identity-operator/finite-rank counterexample in the Hom chapter disproves that extension for the preceding definition (p. 263, PDF p. 269). This local correction does not invalidate the finite-dimensional construction, its universality theorem, or the chapter's examples. The definition and remark were checked as images.

12.5 Exterior algebra: a real change in construction and scope

Xi II §6.4 begins by explicitly assuming characteristic zero and models exterior powers inside antisymmetric tensors (p. 292, PDF p. 298). Its exterior product uses the alternation operator. This model naturally continues the chapter's tensor calculations and geometric language.

Grillet XIII.2–4 builds tensor, symmetric, and exterior algebras by universal properties and quotient ideals over a commutative ring; the exterior relation is $v \otimes v = 0$ (pp. 518–527, PDF pp. 530–539). The basis and alternating-map arguments were sampled in the actual proof pages. This quotient construction is better suited to retaining characteristic 2 and avoiding factorial denominators. In characteristic zero, the models are connected by normalized alternation; normalizations must be translated.

For determinants and minors, Xi I's characterizations and Xi II's geometric motivation remain helpful. The notes choose the quotient exterior construction so the rank and Cauchy–Binet arguments work in arbitrary field characteristic. That is a decision about the desired scope, not an assertion that Xi's stated characteristic-zero treatment is incomplete on its own terms.

12.6 Metric and geometric material is not superseded by module theory

Xi II Chapter 3 develops real and Hermitian inner products, self-adjoint, isometric and normal operators, complexification, orthogonal expansion, projections, least squares, and orthogonal-polynomial

examples. The normal spectral theorem, Theorem 3.62 (pp. 126–128, PDF pp. 132–134), was read with its proof. This supplies an orthonormal conclusion under a positive metric; Jordan form supplies a different conclusion under arbitrary basis changes.

The inspected metric exercises include dual-basis transformations and norm comparisons, providing practice that is not supplied by merely knowing the module structure theorem. Xi's Chapters 4–5 add affine and Euclidean geometry, transformation groups, indefinite spaces, quadrics, and projective geometry. Only their contents and selected contextual passages were audited here, so no claim that a graduate algebra book covers all their proofs or exercises is warranted. Their relevance is already concrete in the Hadamard direction: positive real Gram matrices and exact sign constraints are central, while similarity alone discards the metric problem.

12.7 Xi III, Grillet, and Hall: symmetry with a terminology trap

Xi III has an actual averaging proof for finite complex representations (Theorem 3.20, pp. 109–111, PDF pp. 115–117), followed by examples and remarks about compact groups. Its §3.7 gives dual and tensor representations (pp. 152–162, PDF pp. 158–168). Its module definitions in §4.4 (pp. 186–188, PDF pp. 192–194) connect groups, algebras and actions. The rendered pages show substantive construction coverage.

Grillet IX.7 proves the group-algebra correspondence and Maschke under the field-characteristic condition (pp. 380–384, PDF pp. 392–396). It initially calls a representation "irreducible" when it is **indecomposable** (p. 381, PDF p. 393), explicitly making that correspondence on p. 383. These notes, Xi and Hall use irreducible to mean a nonzero representation with no proper nonzero invariant subspace. Outside a semisimple setting, Grillet IX.7.1 must therefore not be read as an unconditional complete reducibility theorem in the latter terminology. This is a convention difference verified against a rendered page, not an OCR mistake or a false theorem under his definition.

Hall §§4.3–4.7 proves unitary splitting, Schur's lemma, and the $\mathfrak{sl}_2$ classification (pp. 90–101, PDF pp. 103–114). Its unipotent representation of $\mathbb{R}$ demonstrates failure of complete reducibility even over $\mathbb{C}$. Xi III's §4.6 starts from Lie module conditions and explicitly points outside that section for enveloping algebra and PBW theory (pp. 215–216, PDF pp. 221–222). Hall's later PBW, root-system and highest-weight chapters are verified destinations from contents; their full proofs were not compared against Xi's later examples or appendix. Thus "Hall develops a systematic further theory" is supported, while "all Xi representation examples are replaced" is not established.

For Hadamard/Williamson, the immediately useful representation theory is finite cyclic characters and simultaneous splitting of the shift operator. Full compact Lie theory and $\mathfrak{sl}_2$ are enriching branches that can be deferred for that project.

12.8 Homological and geometric graduate books: bridges, not substitutes

Hilton–Stammbach III.1–4 carefully distinguishes extension equivalence from middle-module isomorphism and computes cyclic Ext groups (pp. 84–99, PDF pp. 95–110). Its introduction explains why extensions precede general derived functors. This is the right added construction when invariant splitting fails. Over the field alone, positive Ext/Tor vanish; complex homology remains meaningful. No need for advanced homological machinery has been demonstrated in the sign-lifting constraints of the Hadamard study.

Hartshorne II.5 and II.8 develop local module data and differential universal properties (pp. 108–111, 172–174; PDF pp. 123–126, 187–189). They require sheaves, localization and local rings beyond

one-vector-space arguments. His "A" references make clear that some algebra is cited rather than rederived. Lang I.9 and III.1 add projectivity over Dedekind rings and integral trace duality (pp. 29, 57; PDF pp. 43, 71). They require integral and local arithmetic information that scalar extension forgets.

Hirsch IV.2.2 proves splitting for vector bundles on a paracompact base using partitions of unity (p. 95, PDF p. 105). This is an important extension of the linear splitting discussion, while his preface deliberately chooses not to use tensors or differential forms (printed p. vii, PDF p. 8). An algebraic prerequisite ladder that makes him depend on those subjects would misrepresent the actual book.

The practical decision is selective: use Xi for the concrete, metric, and geometric language; choose Grillet or Hungerford for the module spine; use Mac Lane to sharpen universal reasoning and Hilton–Stammbach when a real exactness obstruction appears. Enter Hall, Lang, Hartshorne, and Hirsch along their particular missing bridges. This is a branching reading plan, not eleven sequential complete readings.

13 Source ledger and page conventions

13.1 Evidence levels

P means the specified text, theorem, construction, or proof was inspected in this task. **V** means a rendered source page was inspected, necessary for scans or a suspected extraction problem. **T** means a contents or introductory description was inspected; it establishes a location and stated scope, not that every theorem or exercise was audited. **C** means a theorem is cited as a standard result and is not fully reproved or source-audited here. The comparison uses these distinctions rather than treating an index hit as proof coverage.

PDF page numbers are **one-based physical page positions in the supplied file**; printed pages are the numbers visible on the book pages. The offsets below apply to the ordinary Arabic-numbered body at the checked locations, not to Roman frontmatter. A section range is a reading destination; a list of inspected pages is the actual evidence, and should not be expanded into a full-book audit.

13.2 Editions and physical files

Key	Supplied source	File pages	Body page conversion
Xi I	席南华, 基础代数 第一卷, Science Press, 2016	207, local metadata	PDF = printed + 6
Xi II	席南华, 基础代数 第二卷, Science Press, 2018	334, local metadata	PDF = printed + 6
Xi III	席南华, 基础代数 第三卷, Science Press, 2021, second printing visible	361, local metadata	PDF = printed + 6
Hungerford	Thomas W. Hungerford, Algebra, Springer edition/reprint of the 1974 text; 1980 Springer preface and 2003 printing note	523, local metadata	PDF = printed + 19

Key	Supplied source	File pages	Body page conversion
Grillet	Pierre Antoine Grillet, Abstract Algebra, second edition, 2007	684, local metadata	PDF = printed + 12
Mac Lane	Saunders Mac Lane, Categories for the Working Mathematician, second edition	331, local metadata	PDF = printed + 13
Hilton–Stammbach	Peter J. Hilton and Urs Stammbach, A Course in Homological Algebra, second edition	376, local metadata	PDF = printed + 11
Hall	Brian C. Hall, Lie Groups, Lie Algebras, and Representations, second edition, 2015	452, local metadata	PDF = printed + 13
Hartshorne	Robin Hartshorne, Algebraic Geometry, Springer	511, local metadata	PDF = printed + 15
Lang	Serge Lang, Algebraic Number Theory, second edition; supplied reprint identifies 1986 edition with later copyright information	372, local metadata	PDF = printed + 14
Hirsch	Morris W. Hirsch, Differential Topology, 1976/reprint	230 reported by Library reader; no local byte audit	PDF = printed + 10 at checked body pages

Local SHA-256 fingerprints and byte sizes for the ten materialized files are saved in `evidence/source-fingerprints.json`. They identify which supplied editions underlie the references; they are not a distribution of source bytes. The original Library filenames, which sometimes have OCR-like punctuation, are not treated as authoritative edition statements.

13.3 Xi I: concrete foundations and determinant characterizations

- **P/T:** title/edition and preface, PDF pp. 3–4; contents, pp. 5–6. The preface connects the course to Kostrikin's framework and names several other references, including Hungerford. This supports pedagogical context, not a claim that the books have identical proof coverage.
- **P:** Chapter 4 opening, printed pp. 75–76 (PDF pp. 81–82): oriented volume, multilinearity, and vanishing on dependent vectors.
- **P:** §4.4, printed p. 93 (PDF p. 99): several characterizations of the determinant, with the preceding exercises including Cauchy–Binet/Laplace identities. This is actual structural content, not merely a chapter title.
- **P/T:** §6.3 opening, printed p. 133 (PDF p. 139): factorization and examples whose irreducibility changes with the coefficient ring.
- **P/T:** §7.4 opening, printed p. 177 (PDF p. 183): elementary symmetric polynomials and invariance under ordering roots.
- **P:** §7.6 opening, printed p. 190 (PDF p. 196): resultant matrix and criterion, with the leading-coefficient degeneracy explicitly visible. The complete resultant proof and exercise collection

were not audited.
- Quotients of sets (§2.4, printed p. 30, PDF p. 36) and matrix-map organization (Chapter 3, starting p. 40, PDF p. 46) were located and sampled.

13.4 Xi II: the principal source comparison

- **P:** §1.4–1.6 samples, printed pp. 16, 19–24 (PDF pp. 22, 25–30): independent subspaces, complements versus quotients, duals and evaluation.
- **P:** §2.4, printed pp. 67–71 (PDF pp. 73–77), selected text: quotient operator and induced-map diagram; characteristic-polynomial factorization; Cayley–Hamilton statement and dual operator discussion.
- **P:** §2.5, printed pp. 72–77 (PDF pp. 78–83), consecutive proof reading: Theorem 2.51 uses Bézout to split annihilated spaces; 2.52–2.53 gives generalized eigenspaces; 2.56–2.57 gives Jordan form; the proof constructs chains by quotient induction and proves uniqueness by dimensions of images of powers.
- **P:** §3.1 opening, printed p. 85 (PDF p. 91), and §3.3 opening, printed p. 109 (PDF p. 115), with examples/exercises from metric linear algebra. **P:** Theorem 3.62 and its normal-operator argument, printed pp. 126–128 (PDF pp. 132–134).
- **T:** Chapters 4–5, printed pp. 161–262 (PDF pp. 167–268): affine and Euclidean geometry, transformations, convexity, indefinite spaces, quadrics, projective geometry. Their complete proofs were not audited; no blanket replacement claim about them is made.
- **P/V:** tensor Definition 6.1, printed p. 263 (PDF p. 269), defines all multilinear scalar functions on the indicated product spaces.
- **P:** tensor Definition 6.11 and Theorem 6.15, printed pp. 273–275 (PDF pp. 279–281): basis construction, uniqueness, and universal property **with a proof**. Subsequent maps, associativity, distributivity, scalar extension, and Kronecker formulas were read on PDF pp. 280–284.
- **P:** §6.3 examples of contraction as trace/composition and structural tensors, printed pp. 280–283 (PDF pp. 286–289).
- **P/V:** Theorem 6.19 and following remark, printed p. 278 (PDF p. 284). The finite-dimensional theorem is correct; the unrestricted infinite-dimensional extension fails for the definition on p. 263. See `VERIFICATION.md` for the explicit counterexample and its scope.
- **P:** §6.4 opening and definition, printed pp. 292–293 (PDF pp. 298–299): characteristic-zero assumption and exterior product via alternation.

13.5 Xi III: selected scanned pages, not unreliable extraction

The Library extraction's earlier 150-page report is not the physical file length. PyMuPDF counted **361 pages**. The scan's extracted text is extremely sparse. The following evidence comes from rendered pages, not OCR guesses.

- **V/T:** contents, PDF pp. 5–6: group actions and structure, one-parameter groups (§2.5, p. 79), matrix Lie groups (§2.6, p. 85), representations (Chapter 3, p. 97), modules (§4.4, p. 186), algebras (§4.5, p. 197), $\mathfrak{sl}_2$ (§4.6, p. 215), Galois theory (Chapter 5, p. 224), and the representations appendix (p. 317).
- **V/P:** §3.2, printed pp. 109–111 (PDF pp. 115–117): complex Hermitian-form convention and Theorem 3.20's explicit finite-group averaging proof, with the compact-group integration comment distinguished from it.
- **V/P:** §3.7, printed pp. 152–153 (PDF pp. 158–159): dual action with inverse, transpose formula, and tensor action.

- **V/P:** §4.4, printed pp. 186–188 (PDF pp. 192–194): module motivation, unital left-module definition, examples, submodules, quotients and maps.
- **V/P:** §4.6, printed pp. 215–216 (PDF pp. 221–222): Lie module conditions and discussion of the enveloping algebra. The page explicitly sends the reader elsewhere for the PBW theory; the notes do not claim that this section proves PBW.
- **V/P:** §5.6, printed pp. 260–262 (PDF pp. 266–268): cyclic multiplicative group, finite-field existence and uniqueness, subfields and Frobenius. This supplies the finite-field prerequisites for the Paley bridge.
- **V/P:** §5.8, printed pp. 272–273, 275–277 (PDF pp. 278–279, 281–283): Noether/Hilbert 90 constructions, norm/trace examples and trace-pairing and coefficient-formula exercises. The trace pairing on p. 277 is an exercise, not a full proof at that location.
- **V/P:** §5.11 opening, printed p. 286 (PDF p. 292): the normal-basis theorem and beginning of its interpolation construction. The rest of that proof, full Galois computations/inverse questions, and the representation appendix remain selective reading destinations, not audited proofs here.

13.6 Graduate sources: where the extra structure is actually built

Grillet. Title/preface and contents were inspected. The detailed samples include VIII.6–7 (printed pp. 336–344, PDF pp. 348–350, 354–356), X.1 and X.3 (pp. 393–395, 401–402; PDF pp. 405–407, 413–414), XI.5–7 (pp. 434–437, 441–443, 447–448; PDF pp. 446–449, 453–455, 459–460), XII.4–5 (pp. 487–489, 493–494; PDF pp. 499–501, 505–506), and XIII.2–4 (pp. 518–526; PDF pp. 530–531, 533–538). These support the module, splitting, balanced tensor, Ext/Tor, and free-algebra comparisons. The general PID proof's key basis and invariant-factor arguments were read; not every ancillary exercise was checked. IX.7, printed pp. 380–385 (PDF pp. 392–397), was inspected for the group-algebra correspondence, Maschke proof, and its nonstandard initial use of "irreducible" for indecomposable. The p. 381 definition was checked as an image. IV.1.6 (printed p. 157, PDF p. 169) was read for cyclic multiplicative subgroups, and V.1.3 (printed pp. 192–193, PDF pp. 204–205) for the finite-field existence/uniqueness proof and field-table exercises. The other IV–VI destinations for norm/trace, orders/valuations and completions are **T**, not a proof audit.

Hungerford. Prefaces, contents, and course suggestions were read, especially printed p. xviii (PDF p. 15), which explicitly places torsion PID modules before VII.1–5 in the proposed linear-algebra course. IV.6 begins at printed p. 218 (PDF p. 237); selected text on torsion/cyclic modules and the structure statements was read at PDF pp. 239–241. VII.4 starts at p. 355 (PDF p. 374), VII.5 at p. 366 (PDF p. 385). The route and theorem locations are verified, but a line-by-line comparison of all proofs in VII was not completed. IV.4 (p. 199, PDF p. 218), IV.5 (p. 207, PDF p. 226), are selective destinations grounded in contents. V.5's finite-field proofs were then read at printed pp. 279–281 (PDF pp. 298–300), including Theorem 5.3, Proposition 5.6, Corollary 5.7, Frobenius, and the explicit exercise constructing fields of orders 9 and 25.

Mac Lane. III.1 (printed pp. 55–58, PDF pp. 68–71), III.2 (pp. 60–62, PDF pp. 73–75), IV.1 (pp. 79–80, PDF pp. 92–93), and IV.2's duality example (pp. 88–89, PDF pp. 101–102) were read. These are actual definitions, examples, and proofs of universality, representability/Yoneda, and adjunction. VIII's abelian-category treatment and later monads/Kan extensions are **T/C** in this study.

Hilton–Stammbach. Edition, contents, and introduction were read; III.1–2 and III.4 were sampled in detail (printed pp. 84–86, 89–90, 97–98; PDF pp. 95–97, 100–101, 108–109). IV.1's definitions and homology construction were read at pp. 117–118 (PDF pp. 128–129). The introduction explicitly motivates the extension-first route before general derived functors. The complete comparison theorem

for resolutions and the full general Ext classification are **C** here; our one-step cyclic cases are proved directly.

Hall. The supplied book is the second edition. §§4.3–4.7 were sampled at printed pp. 90–101 (PDF pp. 103–114), including the unipotent nonsemisimple example, unitary splitting proof, Schur argument, and $\mathfrak{sl}_2$ classification proof. Later root systems, PBW, highest-weight theory, and topology are **T**, not compared proof by proof against Xi.

Hartshorne. Contents and introduction, Chapter II opening, II.5, II.8, and Chapter III opening were read at PDF pp. 9–10, 14–15, 74–75, 123–126, 187–189, 216. In printed pages these include II.5 at pp. 108–111, II.8 at pp. 172–174, and III's opening at p. 201. The algebraic differential universal property and the rational-residue-field cotangent calculation are visible at these locations. The "A" notation distinguishes externally referenced theorems. Affine finite projectivity and general cohomological theorems are cited rather than re-audited in full.

Lang. Title/edition and contents were read. I.1 opening (printed p. 3, PDF p. 17), I.9 Proposition 26 and its local-to-global proof (p. 29, PDF p. 43), and III.1's complementary-module definition/proposition (p. 57, PDF p. 71) were inspected. Full valuation, completion, ramification and class field arguments remain **T/C**. A route to them is identified without claiming that the current notes replace them.

Hirsch. Located by title in Library. Preface/contents, PDF pp. 8–10; Chapter IV.1–2 samples, PDF pp. 96–105 (printed pp. 86–95). The preface's declared avoidance of tensors/forms and IV.2.2's partition-of-unity splitting proof are **P**. Local download via the Library helper returned HTTP 401; no local physical-file metadata or fingerprint is claimed. The read-only Library content route succeeded. The full transversality/Morse proofs are not audited here.

13.7 Existing Hadamard/Williamson research

The reference edition is pinned to repository commit `2834fea31fb4ca7203531ca220e55a2fb7640119`, not a moving branch tip:

- Study and original scope.
- Original constructions and exact sign conventions.
- Second-round structural and spectral proofs.
- Third-round proofs, especially §§1–4, 9, 11–13.
- Third-round limitations.
- Prior independent adversarial review.

Those documents and the relevant exact fixture descriptions were read from Git objects. The new foundation chapter gives its own derivations, identifies where it extends the scope of the earlier prime Paley proof, and retains the distinction between previous audit evidence and newly executed finite checks. Numerical historical/status claims and whole prior benchmark campaigns were not rerun for this foundational task.

14 Verification record, corrections, and limits

14.1 What the evidence means

The notes contain ordinary written proofs and explicitly cited structural theorems. They have undergone a separate self-review for hypotheses, notation, and failure cases, plus exact computational spot checks written independently of the earlier repository search implementations. There is no claim of a

separate human reviewer, Lean/kernel verification, a full-book audit, or a new result about universal Hadamard existence.

`evidence/exact-checks.json` records the actually executed finite checks, software versions and their scope. `evidence/pdf-build.json` and the LaTeX logs record compilation; compilation verifies neither mathematical correctness nor source fidelity. Source fidelity is supported by the specific inspected pages listed in the source ledger, including image inspection where required.

14.2 Central statements checked by written derivation

Statement	Hypotheses and check
Quotient factorization and first isomorphism theorem	Maps, domains, kernels and representative independence are explicit; no complement is assumed
Dual exactness and annihilator identities	Algebraic dual; finite dimension for the double-dual isomorphism and dimension arguments; basis extension for restriction surjectivity
Direct sums and splitting	Idempotent equations proved for modules; vector-space splitting distinguished from module/equivariant splitting; sections form an affine Hom space
Operator/module correspondence	Intertwiner equation checked; finite-dimensional operator modules distinguished from all finitely generated $k[t]$-modules
Canonical forms	Nonzero annihilator obtained without Cayley–Hamilton; Euclidean Smith mechanism explained; arbitrary PID version cited; splitting-field hypothesis retained for Jordan form
Polynomial projectors	Congruences, image/kernel identification, and independence from Bézout representative checked; canonical primary spaces distinguished from chosen cyclic summands
Cyclic Hom and commutant dimensions	Computed from the image of 1 and divisibility; no unjustified extension to two-variable polynomial rings
Hom–tensor formulas	Currying needs no finite dimension; Hom as a tensor needs finite-dimensional source; noncommutative module sides specified
Symmetric/exterior constructions	Quotient universal properties; alternation includes repeated-vector vanishing in characteristic 2; averaging needs invertible factorial
Determinants and minors	Exterior basis proof precedes determinant definition; rank test, Cauchy–Binet and characteristic coefficients derived; determinant of a map between different spaces is a line map
Forms and spectral theorem	Congruence versus similarity; Hermitian convention explicit; positive-definite complex finite-dimensional hypotheses retained; real and indefinite failures exhibited
Ext/Tor and homology	One-step resolution requires a nonzero element of a domain; explicit nonsplit $k[t]$ extension; field Ext/Tor vanishing separated from arbitrary complex homology

Statement	Hypotheses and check
Representations	Maschke needs finite group and invertible order; Schur's scalar result needs algebraic closure and finite dimension; dual action has inverse; Lie tensor action has a sum
Trace pairing	Finite separable extension for nondegeneracy; scalar-extension/CRT proof and purely inseparable counterexample included
Bundles and sheaves	Smooth paracompact bundle splitting distinguished from arbitrary pointwise choices and algebraic sheaf extensions; no category-independent Ext vanishing claim
Hadamard bridge	Integer/real Gram equations; odd finite fields for character sum; exact Paley signs; cyclic Fourier splitting/characteristic hypotheses; compression and phase relaxation retain only necessary implications

The proofs are inspectable in the chapters. The table is a review index, not an independent formal certificate.

14.3 Independently computed finite checks

The script `scripts/check_examples.py` reconstructs fixtures directly and does not import the previous Hadamard project's search or verification code. Its 18 check groups include:

- Polynomial CRT idempotents over $\mathbb{Q}$ and characteristics $2, 3, 5, 7$; a conjugated mixed Jordan fixture checks the actual operator projections.
- Kernel-growth block reconstruction, and the commutant dimension independently computed as the nullity of a 100×100 commutator equation matrix.
- Sixteen cyclic-Hom dimension comparisons using direct intertwiner equations rather than the gcd formula being checked.
- Exterior minors, rectangular Cauchy–Binet, and characteristic coefficients; Kronecker determinant/trace; adjoints for a nonorthonormal Hermitian form.
- Explicit module-section failure over four small prime fields; a nonzero field-complex homology example with chain-homotopy equations; the $\mathfrak{sl}_2$ polynomial commutators for highest weights zero through six.
- Quadratic-character sums and both Paley block constructions for six small primes, plus an actual nonprime-field fixture $\mathbb{F}_9 = \mathbb{F}_3[u]/(u^2 + 1)$ producing order 20.
- Williamson block signs at core three, the core-nine proper-quotient false positive, and all 27 possible normalized symmetric lifts of the **fixed** stronger compressed multiset from the prior audit. None is a Williamson lift. This is a bounded adversarial-fixture replay, not an expanded search campaign.
- The core-seven positive spectral residual with no fourth Williamson row; all 16 two-color-bit equality inputs and the exact Boolean phase identity.

Finite checks can catch sign, transpose, coefficient and characteristic errors in these examples. They do not prove the universal theorems or classify all Hadamard matrices. The infinite-dimensional tensor counterexample is proved by unbounded rank, not established by a finite truncation test.

14.4 Cited results, distinguished from complete proofs here

- The full finitely generated module structure theorem over an arbitrary PID (Grillet VIII.6; Hungerford IV.6). The notes give the Euclidean matrix construction and derive the operator consequences; they do not pretend every PID is Euclidean.
- General derived-functor theory, independence of resolutions, and the complete Ext/extension correspondence (Hilton–Stammbach III–IV; Grillet XII). The cyclic class calculation and its extension interpretation are proved directly in the notes.
- Exhaustion of finite-dimensional complex $\mathfrak{sl}_2$ irreducibles (Hall §4.6). The polynomial models and their irreducibility are proved here; the exhaustion proof was read but is cited rather than reproduced fully.
- Primitive-element theorem for finite separable extensions; finite-field existence; the general commutative-algebra/sheaf correspondence; finite projectivity on affine schemes; implicit-function, partition-of-unity and other geometric prerequisites. Their uses and sources are stated, without implying all are reproved in this study.
- Prior study benchmarks, search completeness records, and historical matrix witnesses remain attributed to their pinned source edition. Only the small fixtures listed above were newly executed. No current worldwide open-problem status or smallest unresolved order is certified.

14.5 Confirmed source correction

Xi II, Theorem 6.19's following remark, printed p. 278 / PDF p. 284. The remark extends the finite-dimensional identification $\mathcal{T}^p_q(V) \cong (V^*)^{\otimes p} \otimes V^{\otimes q}$ to infinite-dimensional spaces. Definition 6.1 on printed p. 263 / PDF p. 269 defines the left side as all multilinear functions on $V^p \times (V^*)^q$. Both pages were rendered and inspected.

For $p = q = 1$, evaluation $(v, \lambda) \mapsto \lambda(v)$ belongs to that function space. If it were a finite tensor sum, there would be $\alpha_i \in V^*$ and $w_i \in V$ such that

$$\lambda(v) = \sum_{i=1}^{r} \alpha_i(v)\lambda(w_i) \qquad \text{for every } v, \lambda.$$

Functionals separate vectors, so $v = \sum_i \alpha_i(v) w_i$ for every v. This puts an infinite-dimensional space in a finite-dimensional span, a contradiction. The corrected statement retains finite-dimensionality, or restricts the multilinear function space to the image of finite tensor sums. The finite-dimensional theorem and the tensor universal property are unaffected. This is one local confirmed overstatement, not a blanket judgment of the text.

14.6 Convention differences and missing hypotheses, not source mistakes

- Grillet IX.7's initial "irreducible" means indecomposable (printed pp. 381, 383; PDF pp. 393, 395). It differs from the simple-module convention in these notes, Xi and Hall. The apparent unconditional complete-reducibility statement must be translated using his definition, not declared false.
- Xi's Hermitian convention is linear in the first argument; these notes use conjugate-linearity in the first. Adjoint and coordinate formulas are translated explicitly.
- Xi II's alternating-tensor model explicitly assumes characteristic zero. Its factorial denominators are valid there; moving the model to arbitrary characteristic without that assumption would be our error, not its theorem.

- A Jordan form needs a splitting minimal polynomial; semisimple $k[t]$-modules need square-free irreducible factors and can lack eigenvectors over the base field. Commuting operators require individual diagonalizability before the simultaneous eigenbasis argument.
- A positive spectral residual and individually feasible compressed rows need not have one common discrete lift. No linear splitting theorem supplies it.
- Fourier eigenvalues over $\mathbb{C}$ are exact algebraic values in the proof; floating-point approximations are not used to certify sign matrices.

14.7 OCR issues and unverified comparisons

Xi III's local PDF has 361 pages, whereas its earlier Library extraction reported 150. Sparse extraction and corrupted symbols are source-access limitations, not mathematical errors. The key cited pages were read as images. Xi I–II also have corrupted extracted signs and indices, so suspected substantive claims were checked against the intended notation and, where material, images. The exact source fingerprints allow later reinspection.

The following comparisons remain deliberately limited: the entire exercises of any two books; Xi II Chapters 4–5 versus a putative replacement; the whole Xi III appendix or Galois computations against Hall/Lang; complete Hungerford VII proof alternatives; and the destination books' full advanced prerequisite coverage. They are not promoted from contents evidence into proven replacement claims. No source book is called filler or wholly replaceable here.

14.8 Remaining access limitation

The exact failed action was local materialization of the Library item titled `Differential Topology (Morris W. Hirsch).pdf` using the provided download helper. It returned `hosted apps tools/list request failed with HTTP status 401`. This is a failed local-download action, not a successful or uncertain transfer and not an automatic sandbox-approval denial. The same source's read-only Library pages were successfully read afterward. Consequently the Hirsch content needed here is sourced, but no physical local PDF page count, byte hash, or full local-PDF audit is claimed.

There is no pending mathematical/source blocker to using the present notes. Completing the optional unverified comparisons would be a new, larger audit; recovering a local Hirsch copy is only necessary for that sort of byte-level or full visual audit.